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Forecasting electricity prices for a day-ahead pool-based electric energy market

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Abstract

This paper considers forecasting techniques to predict the 24 market-clearing prices of a day-ahead electric energy market. The techniques considered include time series analysis, neural networks and wavelets. Within the time series procedures, the techniques considered comprise ARIMA, dynamic regression and transfer function. Extensive analysis is conducted using data from the PJM Interconnection. Relevant conclusions are drawn on the effectiveness and flexibility of any one of the considered techniques. Furthermore, they are exhaustively compared among themselves.

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Keywords: Electricity market; Day-ahead price forecasting; Time series models; Neural networks; Wavelet models

1. Introduction

1.1. Electric energy markets

During the last two decades the electric power industry all over the world has undertaken significant restructuring. In most countries, a cost minimization paradigm has been replaced by a profit maximization one. In the cost minimization framework, a Central Operator (CO) decides centrally how the generating

machines should be operated to minimize total cost while serving all demands. In contrast, in the profit maximization framework, producers, retailers and consumers interact through a market seeking to maximize their respective profits. Two market structures arise commonly in practice: a bilateral contract framework and a pool.

In a bilateral transaction market, any given producer agrees with retailers to supply specified amounts of energy during a contract horizon. Those physical contracts are implemented with the help of an Independent System Operator (ISO) that takes care of the physical requirements needed for the transactions to take place in a secure manner.

In a pool, producers submit to the Market Operator (MO) production bids that typically consist of a set of

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energy blocks and their corresponding minimum selling prices for every hour of the market horizon. Analogously, retailers and large consumers submit to the MO consumption bids that consist of a set of energy blocks and their corresponding maximum buying prices. The MO uses a market-clearing algorithm to clear the market, which results in a market-clearing price as well as the scheduled production and consumption for every hour of the market horizon. The market-clearing price is the price to be paid by retailers and to be charged by producers. Finally, the ISO checks for technical feasibility and, if needed, introduces the minimal required changes to attain a secure operation.

This pool-based electric energy market is the most common arrangement in practice. Sometimes, it coexists with a bilateral contract framework. In this case, the ISO ensures a technical secure operation of the pool plus the simultaneous bilateral contract arrangements. This paper considers the framework above, i.e., a pool that may include bilateral contract arrangements.

1.2. Why forecasting electricity prices?

Producers need to forecast market-clearing prices to respond optimally to the pool and to efficaciously engage in bilateral contracts.

In the short-run, a producer with low capability of altering market-clearing prices (price-taker producer) needs day-ahead price forecasts to optimally self-schedule and to derive its bidding strategy in the pool. In the medium-term, a price-taker producer requires market-clearing price forecasts for several months in order to sign favorable bilateral contracts.

Retailers and large consumers need day-ahead and medium-term market-clearing price estimates for the same reasons as producers. Those price forecasts constitute fundamental information for the retailers (large consumers) to self-schedule and to bid efficiently in the pool; and to engage in profitable bilateral contracts.

1.3. Forecasting framework

Within the framework of a pool-based electric energy market, this paper considers forecasting

techniques to estimate the 24 day-ahead market-clearing prices.

Three families of techniques are considered: time series, neural networks and wavelets. Time series techniques are treated with greater detail because they revealed themselves, through many realistic studies, as the most efficacious tools for day-ahead market-clearing price forecasting. Time series techniques considered include ARIMA, dynamic regression and transfer function.

A naïve but challenging test is used to characterize all forecasting procedures that are analyzed. The 24 market-clearing price forecasts using any technique can be compared to the 24 market-clearing prices of a day *similar* to the one to be forecast. A *similar* day is characterized as follows. A Monday is similar to the Monday of the previous week and the same rule applies for Saturdays and Sundays; analogously, a Tuesday is similar to the Monday of the same week, and the same rule applies for Wednesdays, Thursdays and Fridays. The naïve test is passed if hourly errors for the estimates using any forecasting technique are smaller than the market-clearing prices of the *similar* day. More often than expected, forecasting procedures not carefully tuned up do not pass this test. This naïve test is used in the case studies that are analyzed in this paper.

The time framework to forecast day-ahead market-clearing prices in most markets is explained below and illustrated in Fig. 1. The market-clearing price forecasts for day d are required on day $d-1$, typically at hour h_b (around 10 am). On the other hand, data concerning results for day $d-1$, including market-clearing prices and demands, are available on day $d-2$ at hour h_c (around noon). Therefore, the actual forecasting of market-clearing prices for day d can take place between hour h_c of day $d-2$ and hour h_b of day $d-1$. Additionally, producers and retailers use this period to optimally self-schedule and to produce appropriate bids.

To perform a fair comparison, available data for price prediction is identical for all techniques. To compute price forecasts for hour 1 to 24 of day d , data available to all procedures include price and demand historical data up to hour 24 of day $d-1$, and demand predictions for the 24th hour of day d . The considered historical data spans 53 days.

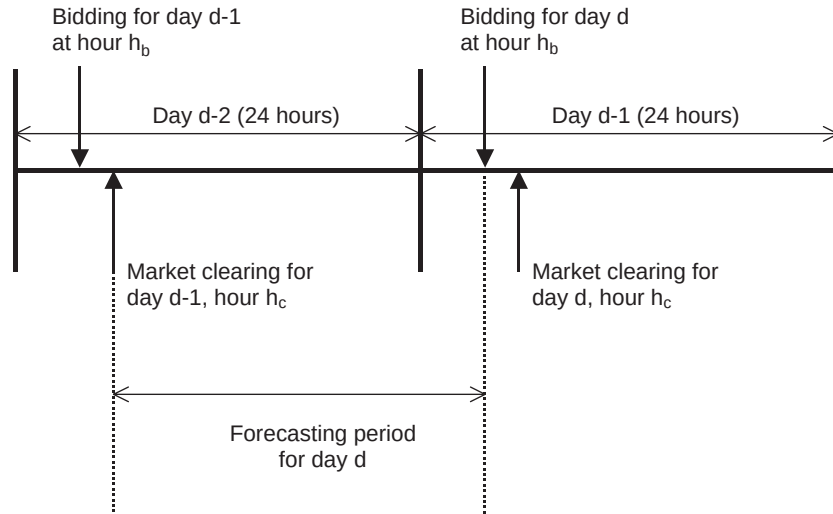


Fig. 1. Time framework to forecast market-clearing prices for day d .

However, this available data is not used in the same way by all techniques.

In the case of ARIMA and neural network models, historical demand data does not significantly improve predictions and therefore it was decided that these models would only rely on price data.

In all techniques the corresponding algorithm uses endogenously computed price forecasts of some hours of day d to forecast prices of future hours of the same day d . For instance, using an ARIMA model to forecast the price of hour 5 of day d , historical price data up to hour 24 of day $d-1$ are used, as well as previously obtained price forecasts for hour, 4, 3 and 2 of day d .

The only explicative variable that has been considered in the study is the demand. The amount of hydro energy stored in reservoirs is a variable of interest for electric energy markets with important hydroelectric producers. However, in this study a market that includes only thermal producers is considered.

Explicative variable data (temperature, hydro resources, thermal equipment outages and the like) for day d , if available, can be easily and fruitfully included in the models presented. Note, however, that unit outage information is generally proprietary and therefore not available to all market agents. Note also that the effect of the temperature, and other weather-related variables, is usually embodied in the demand

forecasts. Nevertheless, for the sake of simplicity and clarity in the comparisons, explicative variables other than demand have not been included in the models presented.

1.4. Time series characterization

In most electricity markets the series of prices presents the following features:

1. High frequency,
2. Non-constant mean and variance,
3. Daily and weekly seasonality,
4. Calendar effect on weekend and holidays,
5. High volatility,
6. Presence of outliers.

The presented models are selected once these characteristics are inspected and understood.

These characteristics can be observed in Fig. 2 for the price series of the PJM Interconnection (PJM, 2003) for the year 2002. In this figure, the actual hourly price is the demand-weighted average of all day-ahead locational marginal prices in that hour for the whole PJM interconnection. These prices are available from PJM (2003).

The demand for electricity partly explains price behavior. Fig. 3 depicts hourly electricity demand for the year 2002 in the PJM Interconnection.

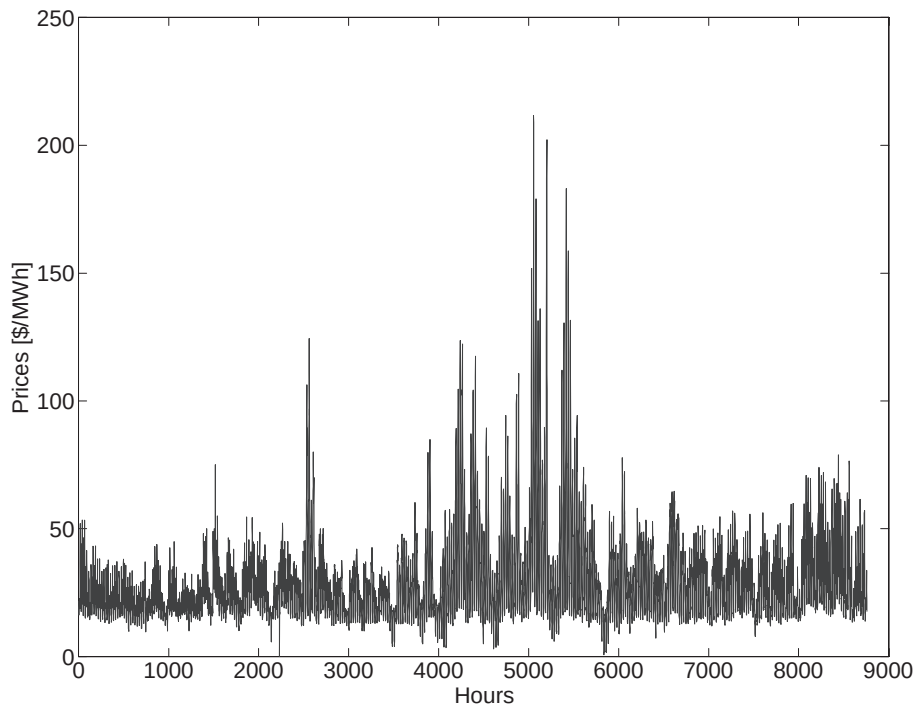


Fig. 2. Electricity price series for year 2002 in the PJM Interconnection.

It should be noted that as a result of the possibly irrational bidding behavior by market agents (producers and consumers), price series are more volatile than demand series.

The non-constant mean feature of the price series is alleviated by differentiating the original series using factors of the form $(1-B^s)$, where B is the back-shift operator and s can be 1 (hourly differentiation), 24 (daily differentiation), 168 (weekly differentiation), or other values depending on the series. The non-constant variance is alleviated by taking logarithms. Fig. 4 shows the resulting differentiated logarithmic price series. It can be observed that this series has relatively constant mean and variance.

Daily and weekly seasonalities are typically taken into account through the use of seasonality models of orders 24 and 168, respectively. The calendar effect is taken into account by incorporating ad-hoc logic.

The high frequency and high volatility features are characteristics inherent to the series that cannot be changed.

Outliers have not been explicitly treated. However, the automatic handling of outliers provided by the

estimation software used (SCA, 2003) did not improve predictions in the case studies analyzed.

1.5. Overview of forecasting procedures

The three time series models presented constitute a class of stochastic processes used to analyze time series, and they share a common methodology due to Box, Jenkins, and Reinsel (1994). The analysis is based upon different probability models to represent the data, including the explanatory variables. The three techniques differ when modeling the relationship between prices and errors. ARIMA relates current prices to past prices, and current errors to previous errors; dynamic regression relates current and past prices and demands; and transfer function relates current prices to past prices, demands and errors.

Artificial Neural Networks (ANNs) are mathematical models that resemble the functioning of the human brain. ANNs need to be trained with a set of inputs to produce the specified outputs. They are composed of an input layer, one or more hidden layers and an output layer. When forecasting time series,

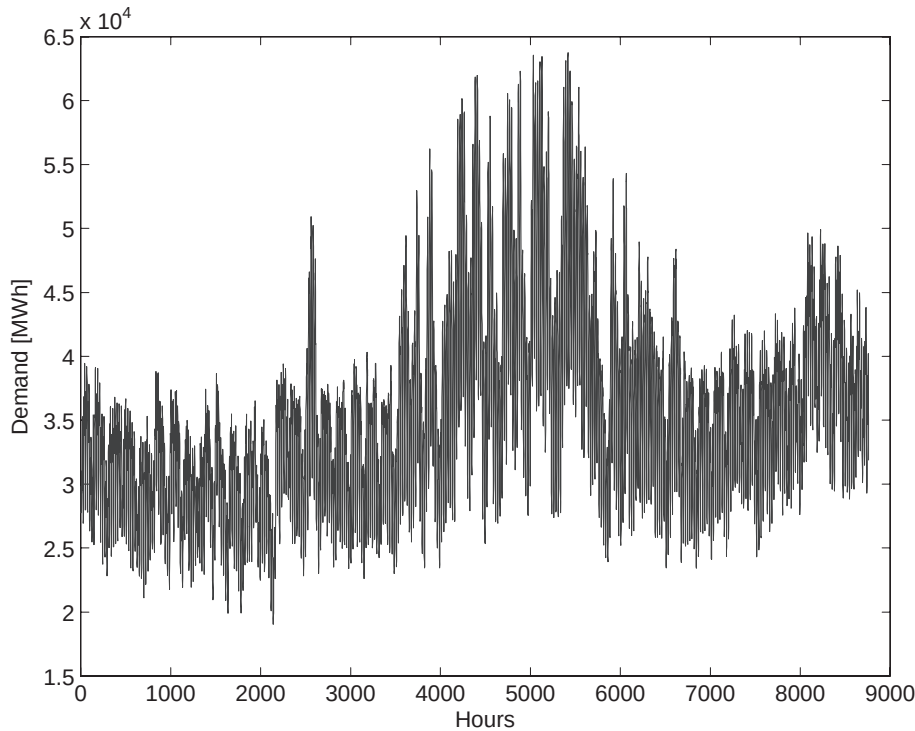


Fig. 3. Electricity demand series for year 2002 in the PJM Interconnection.

ANNs are interesting because they can approximate any nonlinear function. In particular, feedforward backpropagation neural networks are specially suited to forecasting electricity prices because they can process nonlinearities using sigmoid functions for the inputs and linear functions for the outputs.

Wavelet Transform Analysis (WTA) has been extensively used for various signal processing applications in the past, but its use in predicting electricity prices is recent. It is an alternative to the classical Fourier Transform (FT) for analyzing non-stationary series. The basic difference between the FT and WTA is that WTA uses short windows at high frequencies and long windows at low frequencies, in contrast to FT, which uses a single window.

1.6. Literature review and paper contributions

Regarding time series analysis, the following references are relevant: dynamic regression and transfer function papers include [Nogales, Contreras, Conejo, and Espínola \(2002\)](#); ARMA papers include

[Fosso, Gjelsvik, Haugstad, Birger, and Wangensteen \(1999\)](#); and ARIMA papers include [Contreras, Espínola, Nogales, and Conejo \(2003\)](#) in the market of mainland Spain.

Artificial intelligence techniques, such as ANNs, have been applied to forecasting prices in the England–Wales pool ([Ramsay & Wang, 1998](#)), the Australian market ([Szkuta, Sanabria, & Dillon, 1999](#)), the PJM Interconnection ([Hong & Hsiao, 2002](#)), and the New England ISO ([Zhang, Luh, & Kasiviswanathan, 2003](#)).

In addition, wavelet transform analysis has been used in the England–Wales pool ([Kim, Yu, & Song, 2002](#)).

Relevant references not directly related to electricity price forecasting follow. References on gas and oil price forecasting include [Morana \(2001\)](#), [Cabedo and Moya \(2003\)](#) and [Buchanan, Hodges, and Theis \(2001\)](#). Electricity demand is forecast in [Darbellay and Slama \(2000\)](#) and [Harris and Liu \(1993\)](#). Residential gas consumption forecasting is analyzed in [Liu and Lin \(1991\)](#) and forecasting in oil markets is treated in [Abramson and Finizza \(1995\)](#).

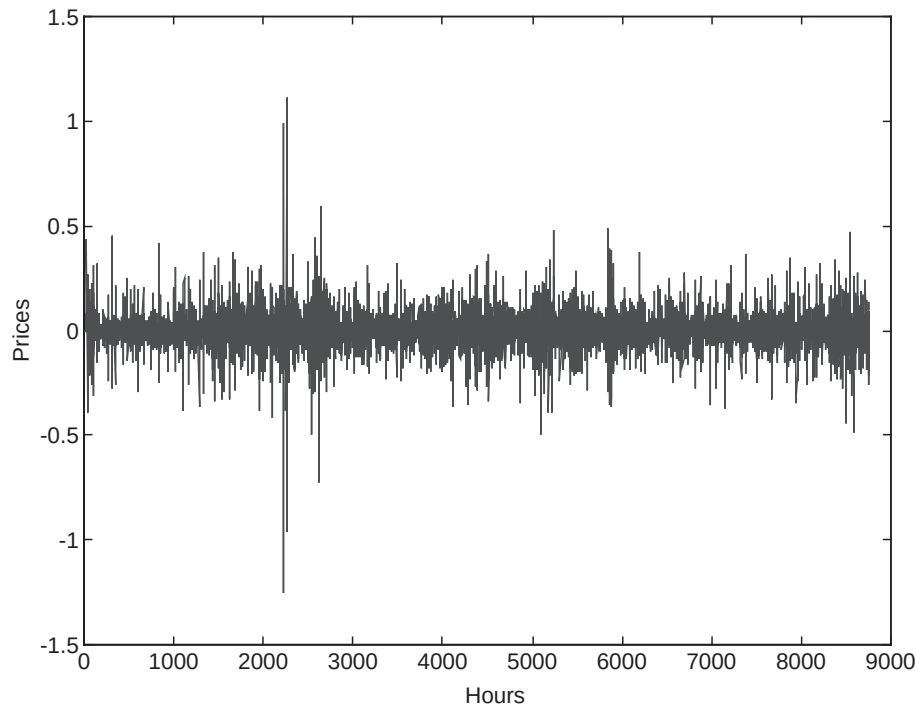


Fig. 4. Logarithmic and differentiated (1 and 24) price series for year 2002 in the PJM Interconnection.

The main contributions of this paper are:

1. Time series algorithms, including dynamic regression, transfer function and ARIMA are developed to forecast day-ahead electricity prices. The building of these models is based on a thorough analysis of historical price and demand data.
2. Two unconventional techniques, a neural network algorithm and a wavelet transform procedure, are proposed to predict day-ahead prices. The neural network procedure relies on precise optimization algorithms.
3. All prediction techniques are compared using a realistic case study based on electricity price data from the PJM Interconnection corresponding to year 2002. Appropriate conclusions are drawn.

1.7. Paper organization

This paper is organized as follows. In Section 2, the three families of forecasting procedures are described. As regards time series analysis, the techniques

considered include ARIMA, dynamic regression and transfer function. Additionally, procedures based on neural networks and wavelets are analyzed. In Section 3, a case study based on data from the PJM Interconnection for the year 2002 is analyzed in detail. Section 4 provides some relevant conclusions.

2. Procedures

2.1. Time series models: ARIMA, Dynamic Regression and Transfer Function

In this section, three models based on time series analysis are presented: ARIMA, Dynamic Regression (DR) and Transfer Function (TF).

These models have been selected following a careful analysis of the main features of the price series. It should be noted that linear regression models cannot be used due to the presence of serial correlation in the errors. Therefore, it is necessary to use models that can handle correlated errors, such as the aforementioned models.

The description of the standard statistical methodology to construct a model is presented below:

- Step 0. A class of models is formulated assuming certain hypotheses.
- Step 1. A particular model is identified for the series being considered.
- Step 2. The parameters of the model are estimated.
- Step 3. If the hypotheses of the model are validated, the procedure continues in Step 4; otherwise the procedure continues in Step 1 to refine the model.
- Step 4. The model is used to forecast.

These steps are explained below for the three time series models considered.

2.1.1. Step 0: Model selection

2.1.1.1. ARIMA model. The proposed general ARIMA model has the form

$$\phi(B)p_h = c + \theta(B)\varepsilon_h \quad (1)$$

where p_h is the price at hour h and ε_h is the error term. $\phi(B)$ and $\theta(B)$ are polynomial functions of the back-shift operator B (observe that $B^s p_h = p_{h-s}$). That is

$$\phi(B) = 1 - \sum_{k=1}^{n_F} \phi_k B^k \quad (2)$$

where $\phi(B)p_h = 1 - \phi_1 p_{h-1} - \phi_2 p_{h-2} - \dots - \phi_{n_F} p_{h-n_F}$, and ϕ_k ($k=1, \dots, n_F$) are polynomial coefficients, and

$$\theta(B) = 1 - \sum_{k=1}^{n_T} \theta_k B^k \quad (3)$$

where $\theta(B)\varepsilon_h = 1 - \theta_1 \varepsilon_{h-1} - \theta_2 \varepsilon_{h-2} - \dots - \theta_{n_T} \varepsilon_{h-n_T}$, and θ_k ($k=1, \dots, n_T$) are polynomial coefficients.

The number of terms of the polynomial functions $\phi(B)$ and $\theta(B)$, n_F and n_T , respectively, depends on the time series under analysis.

Note that including in the left hand side of expression (1) factors of the form $(1-B^s)$ allows appropriately taking into account for the seasonality effects.

Also, note that model (1) relates actual prices to past prices through function $\phi(B)$, and actual errors to past errors through function $\theta(B)$. Finally, certain hypotheses on the error terms, ε_h , are needed to ensure

the effectiveness of the predictions. If the model is properly selected, these errors should behave as white noise.

2.1.1.2. Dynamic regression model. In this model, the price at hour h is related to the values of past prices and past demands. This is carried out to obtain a model that has uncorrelated errors. The general dynamic regression model has the form

$$p_h = c + u(B)p_h + v(B)d_h + \varepsilon_h \quad (4)$$

where p_h is the price in hour h , d_h is the demand in hour h , c is a constant and ε_h is the error term. Polynomial functions of the back-shift operator B , $u(B)$ and $v(B)$, have the form

$$u(B) = \sum_{k=1}^{n_U} u_k B^k \quad (5)$$

$$v(B) = \sum_{k=1}^{n_V} v_k B^k \quad (6)$$

and the constants u_k ($k=1, \dots, n_U$) and v_k ($k=1, \dots, n_V$) are polynomial coefficients that should be selected and estimated to achieve white noise errors.

2.1.1.3. Transfer function model. Both price and demand series are assumed to be stationary in this model (i.e., they have constant mean and variance). The general form of this model is

$$p_h = c + w(B)d_h + N_h \quad (7)$$

where c is a constant, p_h is the price at hour h , d_h is the demand at hour h , and $w(B) = \sum_{k=1}^{n_{TF}} w_k B^k$ is a polynomial function of the back-shift operator (the same polynomial structure as in the DR model). N_h is a disturbance term that follows an ARMA model of the form

$$N_h = \frac{\theta(B)}{\phi(B)} \varepsilon_h \quad (8)$$

where $\phi(B)$ and $\theta(B)$ are already defined polynomial functions of the back-shift operator. Finally, ε_h is the error term that is assumed to be a white noise.

Model (7) relates actual prices to past demands through function $w(B)$, actual prices to past errors through function $\theta(B)$, and actual prices to past prices through function $\phi(B)$.

It should be noted that the proposed transfer function model is a finite order approximation of the proposed dynamic regression model provided that $1-u(B)$ is invertible.

2.1.2. Step 1: Model identification

This step is common for the three time series models considered. Its target is to identify which polynomial parameters should be estimated, because they affect the forecasting. The initial selection is based on the observation of the autocorrelation and partial autocorrelation plots. Further refinement of the selection is based on physical knowledge and on engineering judgment.

In this step, it might be convenient to make the time series stationary (constant mean and variance). To that end, a transformation of the original price data may be necessary. A logarithmic transformation is usually applied to the price data to attain a more stable variance, and factors of the form $(1-B^s)$ can be used to attain a more stable mean. Superscript s can be 1 (hourly differentiation) and/or 24 (daily differentiation) and/or 168 (weekly differentiation), depending on the actual data.

2.1.3. Step 2: Polynomial parameter estimation

Once the parameters of the polynomials different from 0 have been identified (through plot observation, physical knowledge and engineering judgment), these parameters should be estimated. The estimation procedure is based on available historical data. Good estimators are found assuming that the data constitute observations of a stationary time series and maximizing the likelihood function with respect to the polynomial parameters.

Sometimes, the series contain unusual observations, or outliers. In these cases, it is appropriate to use a procedure to detect and minimize the effect of these outliers. With this adjustment, better forecasting performance is usually achieved.

Conventional and commercially available software such as SAS (SAS, 2003) or SCA (SCA, 2003) can be used to carry out the actual estimation.

2.1.4. Step 3: Validation of model hypotheses

In this step, a diagnosis check is used to validate the model assumptions.

If the estimated model is appropriate, then the residuals (actual prices minus predicted prices) should behave in a manner consistent with the model. Residuals must satisfy the requirements of a white noise process: zero mean, constant variance, zero correlation and normal distribution. Taking tests for randomness, such as the one based on the Ljung-Box statistics, Box et al. (1994), and observing plots, such as the autocorrelation and partial autocorrelation plots, allow checking of these requirements.

If the hypotheses on the residuals are validated, then the corresponding model can be used to forecast prices and this step is concluded successfully. Otherwise, the residuals contain a certain structure that should be analyzed to refine the model, and the procedure continues back in Step 1. To refine the model a careful inspection of the autocorrelation plots of the residuals should be performed.

2.1.5. Step 4: Actual prediction

In this step, the corresponding model from Step 2 is used to predict future values of prices, typically 24 h ahead. It should be noted that prediction quality deteriorates as the predicted hour increases, i.e., the error of the estimate of hour 24 is typically greater than the error of the estimate of hour 1.

Conventional and commercially available software such as SAS (2003) or SCA (2003) can be used to carry out the actual prediction in a convenient manner.

2.2. Neural networks

In this section a Neural Network model is proposed to predict prices.

The architecture of the network is important. If the transfer functions and input variables are selected appropriately, a good prediction can be obtained. Different types of network architectures can be used, but only the multilayer perceptron with one hidden layer is considered in this paper. In this architecture neurons are organized in the form of layers. It includes an input layer of source nodes that communicates with a hidden layer, whose computation nodes are denominated hidden neurons. The outputs of the hidden layer translate the input to the output layer of neurons. Therefore, this network works in a directed manner (see Haykin, 1999). The neural network considered is fully connected in the

sense that every node belonging to each layer is connected to every other node belonging to the adjacent forward layer. Moreover, the information flow progresses from left to right, i.e., from the input to the output layers through the hidden layers. This is depicted in Fig. 5.

The general form of the output of the neural network considered is the following

$$p_h^{\text{est}} = \sum_{j=1}^{n_H} a_j f_j \left(\sum_{i=1}^{n_I} b_{ji} p_{ih}^* + b_{j0} \right) + a_0 \quad (9)$$

where p_{ih}^* ($i=1, \dots, n_I$) are past prices required to predict the price in hour h , p_h ; p_h^{est} is the predicted price for hour h , n_H is the number of hidden neurons and n_I the number of input data. Functions $f_j(\cdot)$ ($j=1, \dots, n_H$) are normally hyperbolic tangent functions, i.e.,

$$f_j(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}; j = 1, \dots, n_H \quad (10)$$

The typical number of inputs necessary to achieve a good estimation in the simulations performed is

$n_I=19$, which is equal to the number of terms typically required by the transfer function model.

An optimization technique is used to train the network, i.e., to estimate the network parameters a_j ($j=1, \dots, n_H$) and b_{ji} ($j=1, \dots, n_H$; $i=1, \dots, n_I$). This technique uses the solver CONOPT under GAMS, Brooke, Kendrick, Meeraus, and Raman (1998).

To carry out the training, known outputs are compared with those generated by the network, and errors are computed. Then these errors are minimized in the minimum squared sense, which allows the estimation of the weighting parameters a_j ($j=1, \dots, n_H$) and b_{ji} ($j=1, \dots, n_H$; $i=1, \dots, n_I$). That is, these parameters are estimated through the solution of the problem below:

$$\begin{aligned} &\text{minimize} \quad \sum_{h=1}^{n_T} [p_h^{\text{true}} - p_h^{\text{est}}(a_j, b_{ji})]^2 \\ &\text{subject to} \quad a_j^{\text{low}} \leq a_j \leq a_j^{\text{up}}, j = 1, \dots, n_H \\ &\quad \quad \quad b_{ji}^{\text{low}} \leq b_{ji} \leq b_{ji}^{\text{up}}, j = 1, \dots, n_H; i = 1, \dots, n_I \end{aligned} \quad (11)$$

where the optimization variables are the parameters a_j ($j=1, \dots, n_H$) and b_{ji} ($j=1, \dots, n_H$; $i=1, \dots, n_I$), n_T is the

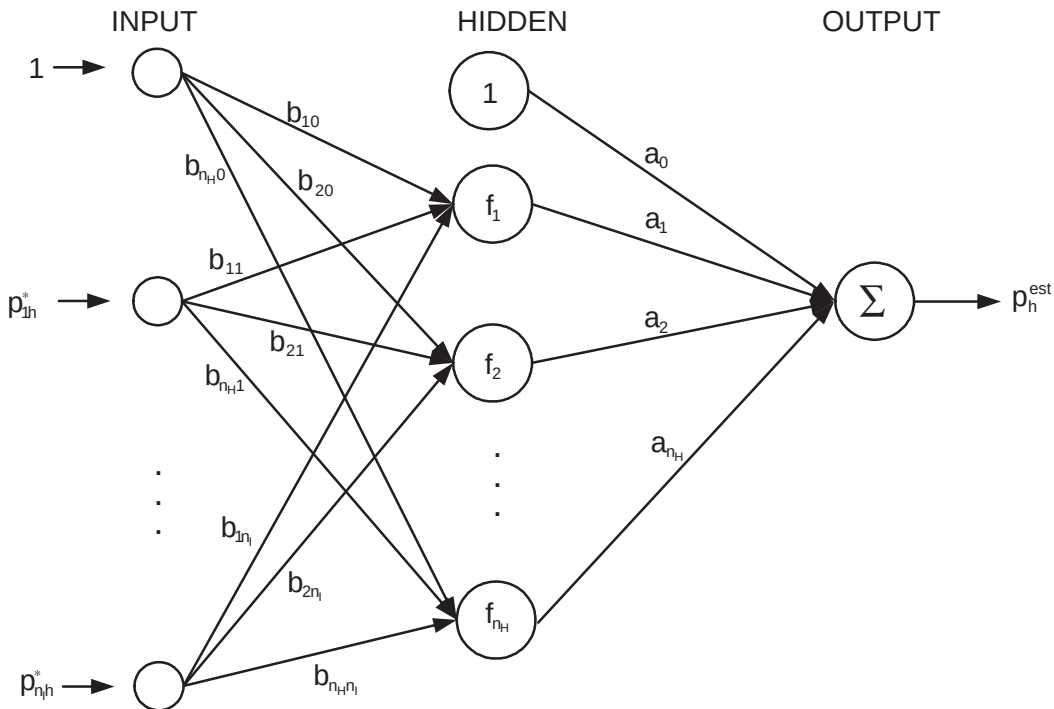


Fig. 5. Neural network architecture used for training and prediction.

number of true prices used for training, p_h^{true} the true price of hour h , and $p_h^{\text{est}}(\cdot)$ the estimate price for hour h . Superscripts “low” and “up” indicate lower and upper bounds, respectively. To avoid local minima, the optimization is initialized from different initial solutions and rerun. The solution that achieves the smallest error is used for forecasting. Training is performed once per day to use the 24 true prices available every new day. That is, parameters a_j ($j=1, \dots, n_H$) and b_{ji} ($j=1, \dots, n_H$; $i=1, \dots, n_I$) are recalculated solving (11) once a day. All data used for training are true values.

Once the neural network is trained and its parameters estimated, it is used to predict hourly prices on a hour by hour basis. That is, to predict the price of hour h , the network is evaluated using the n_I input prices that it requires, p_{ih}^* ($i=1, \dots, n_I$). These prices are generally true prices, but for the last hours to be forecast the complete set of true prices required might not be available; in this case estimates are used to complete the set. The * superscript indicates that

the input prices may be true or estimated. The network evaluation using (9) is repeated for the 24 h of day d . This forecasting procedure is illustrated in Fig. 6.

2.3. Wavelet prediction

This section describes a wavelet transform model for forecasting day-ahead electricity prices.

Prediction techniques based on wavelets (see Kim et al., 2002) make it possible to take into account the time dependency of prices and the correlations with other relevant series, such as demand. The wavelet transform is similar to the Fourier transform, but it can be used with time series that are not periodic. The wavelet transform basically works as follows:

1. Select a typical wavelet function and the origin of the series to be analyzed. This function is used to project the series under analysis (price or demand).

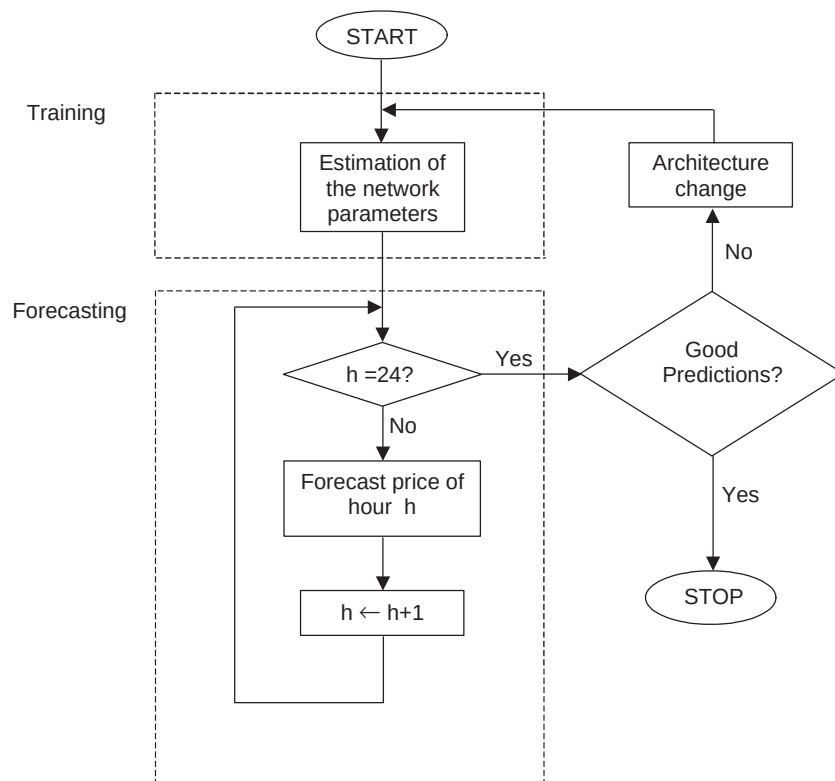


Fig. 6. Training and forecasting using a neural network.

2. Project the series on the wavelet function and obtain the decomposition coefficient that provides the degree of similitude of the price series and the wavelet function.
3. Shift the wavelet function forward and repeat steps 1 to 2 until the original series has been fully analyzed.
4. Scale the wavelet function appropriately and repeat steps 1 to 3. This allows generation of a new set of decomposition coefficients for a lower resolution level.
5. Repeat Step 4 to consider all required levels of resolution.

The wavelet transform procedure is carried out considering a finite number of positions and resolution levels (discrete wavelet transform), usually through the use of dyadic intervals, see Mallat (1988). Using this technique, the decomposition coefficients of the wavelet transform of the hourly price series are given by

$$\begin{aligned}
 p_{mn}^W &= 2^{-\left(\frac{m}{2}\right)} \sum_{h=0}^{H-1} p_h w\left(\frac{h - n \times 2^m}{2^m}\right) \\
 &= 2^{-\left(\frac{m}{2}\right)} \sum_{h=0}^{H-1} p_h w_{mn}(h)
 \end{aligned} \quad (12)$$

where $w(\cdot)$ is the selected wavelet function, p_h is the value of the price at hour h , H is the length of the series, p_{mn}^W is the decomposition coefficient corresponding to position n and resolution level m . Note that if the number of observations, H , is divisible by 2^m , then the number of coefficients at each resolution level is $H/2^m$. To speed up calculations, expression (12) can be treated as a convolution, and the efficient Fast Fourier Transform used, as stated in Frazier (1999).

An efficacious manner to select the wavelet functions is the multi-resolution technique proposed by Mallat (1988) based on using a father wavelet function and its complement, a mother wavelet function. The father function allows the low frequency components of the series to be derived, while the mother one allows components of high frequency to be derived. Additionally, it is convenient to choose orthogonal wavelet functions due to their appropriate mathematical properties, see Daubechies (1988).

The so-called “approximation series”, A_m^P ($m=1, \dots, M$), and the “detail series”, D_m^P ($m=1, \dots, M$), are defined as

$$A_m^P = \sum_n p_{mn}^\Phi \varphi_{mn}(h); m = 1, \dots, M \quad (13)$$

and

$$D_m^P = \sum_n p_{mn}^\Psi \psi_{mn}(h); m = 1, \dots, M \quad (14)$$

where $\varphi_{mn}(h)$ and $\psi_{mn}(h)$ are the father and mother wavelet functions, and p_{mn}^Φ and p_{mn}^Ψ are the coefficients obtained through Eq. (12). Note that A_m^P ($m=1, \dots, M$) and D_m^P (p) are series, i.e., $A_m^P = \{A_{m1}^P, \dots, A_{mH}^P\}$ and $D_m^P = \{D_{m1}^P, \dots, D_{mH}^P\}$.

It can be shown, see Mallat (1988) and Frazier (1999), that the original price series $p = \{p_1, \dots, p_H\}$ can now be reconstructed by

$$p = D_1^P + \dots + D_M^P + A_M^P \quad (15)$$

which is the denominated multiresolution decomposition of the price series p .

The main idea of the method proposed is to examine the relationship between price and demand considering that the variation in each variable has been restricted to a specific scale. The forecasting procedure using the wavelet transform is as follows.

2.3.1. Step 1. Mitigation of the effect of outliers

Outliers are detected and their pernicious effects attenuated using an appropriate mitigation technique, see Peña, Tiao, and Tsay (2001). However, outlier mitigation is outside the scope of this paper.

2.3.2. Step 2. Derivation of the approximation and detail series

Using the available price and demand series up to and including day $d-1$, decompose the considered series of prices and demands considering M resolution levels in order to obtain the approximation and detail series A_M^D and D_i^D ($i=1, \dots, M$); and A_M^P and D_i^P ($i=1, \dots, M$) for the true demand and price series

$$p^{\text{true}} = A_M^P + \sum_{i=1}^M D_i^P \quad (16)$$

and

$$d^{\text{true}} = A_M^D + \sum_{i=1}^M D_i^D \quad (17)$$

2.3.3. Step 3. Derivation of correlation coefficients

Using the approximation and detail series obtained in Step 2, determine the relationship between demand and price series computing the coefficients of the linear multivariate regression model below

$$A_M^P = \alpha_M^A + \beta_M^A A_M^D + \gamma_M^A (A_M^D)^2 + \varepsilon_M^A \quad (18)$$

and

$$D_m^P = \alpha_m^D + \beta_m^D D_m^D + \gamma_m^D (D_m^D)^2 + \varepsilon_m^D; \quad m = 1, \dots, M \quad (19)$$

The regression coefficients obtained in this step are α_M^A , β_M^A , γ_M^A , and α_m^D , β_m^D , γ_m^D , $m=1, \dots, M$.

2.3.4. Step 4. Determination of the approximation and the detail functions through the correlation coefficients obtained in Step 3

The demand forecast series for day d , d^{est} , which is considered known, is decomposed using

$$d^{\text{est}} = A_M^{D \text{ est}} + \sum_{i=1}^M D_i^{D \text{ est}} \quad (20)$$

and the corresponding approximation and detail series for day d , $A_M^{D \text{ est}}$, and $D_m^{D \text{ est}}$, $m=1, \dots, M$, determined. Next, these series and the linear regression coefficients obtained in Step 3 are used to estimate the approximation and detail series related to prices for day d , i.e.,

$$A_M^{P \text{ est}} = \alpha_M^A + \beta_M^A A_M^{D \text{ est}} + \gamma_M^A (A_M^{D \text{ est}})^2 \quad (21)$$

and

$$D_m^{P \text{ est}} = \alpha_m^D + \beta_m^D D_m^{D \text{ est}} + \gamma_m^D (D_m^{D \text{ est}})^2; \quad m = 1, \dots, M \quad (22)$$

2.3.5. Step 5. Price estimation

The price for day d ($h=1, \dots, 24$) is obtained by adding the approximation and detail series derived in Step 4,

$$p^{\text{est}} = A_M^{P \text{ est}} + \sum_{i=1}^M D_i^{P \text{ est}} \quad (23)$$

Some final comments are appropriate. First, Daubechies wavelets are the most commonly used wavelets in applications. For these families of wavelets, the smoothness increases as the order of the functions do; nevertheless, the support intervals also increase, which may cause the prediction to deteriorate. Therefore, low order wavelet functions are generally

advisable. Second, a long estimation period may cause inaccuracies because market conditions evolve with time. On the other hand, a short estimation period may cause volatile estimations. Therefore, an appropriate estimation period should be selected based on trial and error. Finally, it should be noted that efficient numerical implementations of the procedure above are readily available (see, e.g., [The MathWorks, 2003](#)).

2.4. Prediction accuracy

To assess the prediction capacity of all proposed models, different statistical measures are used. This capacity can be checked afterwards once the true market prices are available. For all the weeks under study, two type of average prediction errors are computed: the one corresponding to the 24 h of each day and the one corresponding to the 168 h of each week.

The daily error is computed as

$$e_{\text{day}} = \frac{1}{24} \sum_{h=1}^{24} \frac{|p_h^{\text{true}} - p_h^{\text{est}}|}{\bar{p}_h^{\text{true}, 24}} \quad (24)$$

where

$$\bar{p}_h^{\text{true}, 24} = \frac{1}{24} \sum_{h=1}^{24} p_h^{\text{true}} \quad (25)$$

where e_{day} is the daily error, p_h^{true} is the actual price in hour h , p_h^{est} is the predicted price for that hour, and $\bar{p}_h^{\text{true}, 24}$ is the average true price for the day.

The denominator of the right hand side of (24) is the average daily price to avoid the adverse effect of prices close to zero.

Peak and off-peak daily errors can be computed as

$$e_{\text{peak}} = \frac{\left| \sum_{h \in \{7-13, 18-21\}} p_h^{\text{true}} - \sum_{h \in \{7-13, 18-21\}} p_h^{\text{est}} \right|}{\sum_{h \in \{7-13, 18-21\}} p_h^{\text{true}}} \quad (26)$$

$$e_{\text{off-peak}} = \frac{\left| \sum_{h \in \{1-6, 14-17, 22-24\}} p_h^{\text{true}} - \sum_{h \in \{1-6, 14-17, 22-24\}} p_h^{\text{est}} \right|}{\sum_{h \in \{1-6, 14-17, 22-24\}} p_h^{\text{true}}} \quad (27)$$

Table 1
Neural network weighting parameters for February 18, 2002

b_{ji}												
$j=$	0	1	2	3	4	5	6	7	8	9	10	
$i=$	0	–	–0.14	–0.30	0.01	0.30	–0.30	–0.30	–0.28	–0.30	0.30	–0.30
	1	–	–0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
	2	–	–0.30	–0.30	0.30	–0.30	–0.24	–0.30	0.21	–0.30	0.30	0.30
	3	–	–0.30	–0.30	–0.30	–0.30	0.30	–0.30	0.30	–0.09	0.30	0.30
	4	–	0.30	0.30	0.30	0.10	–0.30	0.30	0.30	0.30	0.16	0.30
	5	–	–0.30	0.30	–0.30	–0.30	–0.30	0.30	–0.30	–0.30	–0.30	–0.30
	6	–	0.30	–0.22	0.24	0.30	–0.30	–0.21	0.30	0.30	–0.30	0.14
	7	–	–0.30	0.30	–0.30	–0.30	–0.30	0.30	0.26	0.30	–0.30	–0.30
	8	–	0.03	0.22	–0.30	0.30	0.30	0.12	0.30	–0.30	–0.28	–0.30
	9	–	–0.30	–0.3	–0.30	–0.27	0.24	–0.30	0.30	0.30	0.30	–0.30
	10	–	0.30	–0.3	0.05	0.30	0.30	–0.07	0.30	–0.30	–0.30	0.16
	11	–	–0.15	0.18	–0.30	0.30	–0.30	0.02	0.19	–0.30	0.30	–0.30
	12	–	0.30	0.30	–0.30	0.30	0.30	0.30	–0.23	–0.30	–0.30	–0.02
	13	–	0.10	–0.11	–0.30	0.30	–0.30	0.01	–0.30	0.30	0.30	0.13
	14	–	–0.05	0.30	0.30	0.28	0.30	0.30	–0.30	–0.30	–0.30	0.30
	15	–	0.05	–0.28	–0.30	–0.30	0.30	–0.30	–0.30	0.03	0.30	0.30
	16	–	–0.30	0.30	–0.04	–0.30	0.30	0.30	–0.30	–0.30	0.30	0.30
	17	–	0.00	–0.06	–0.30	–0.30	–0.30	–0.15	–0.30	0.23	0.30	0.30
	18	–	–0.30	0.30	0.30	–0.30	–0.30	0.30	–0.19	0.30	–0.04	–0.30
	19	–	0.02	0.30	–0.30	–0.30	–0.30	0.30	0.28	0.30	–0.30	–0.30
a_j	0.3	–0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3

These errors are of interest to effectively define bilateral contracts between electricity producers and consumers.

Analogous to the daily error, the weekly error, e_{week} , is computed as

$$e_{\text{week}} = \frac{1}{168} \sum_{h=1}^{168} \frac{|p_h^{\text{true}} - p_h^{\text{est}}|}{\bar{p}_h^{\text{true}, 168}} \quad (28)$$

where

$$\bar{p}_h^{\text{true}, 168} = \frac{1}{168} \sum_{h=1}^{168} p_h^{\text{true}} \quad (29)$$

The denominator of the right hand side of (28) is the average weekly price to avoid the adverse effect of prices close to zero.

Additionally, the forecast mean square error, e_{FMSE} , is computed. This parameter is the square root of the average of either the 24 (daily) or the 168

(weekly) square differences between the predicted prices and the actual ones, i.e.,

$$e_{\text{FMSE}}^{\text{day}} = \sqrt{\frac{1}{24} \sum_{h=1}^{24} (p_h^{\text{true}} - p_h^{\text{est}})^2},$$

$$e_{\text{FMSE}}^{\text{week}} = \sqrt{\frac{1}{168} \sum_{h=1}^{168} (p_h^{\text{true}} - p_h^{\text{est}})^2} \quad (30)$$

An index of the uncertainty of a model is the variability of what is still unexplained after fitting the model, which can be measured through the estimation of the variance of the error term. The smaller the variance, the more precise is the prediction of prices.

Consistent with definitions (24) and (28), daily and weekly error variances can be estimated as

$$\sigma_{e, \text{day}}^2 = \frac{1}{24} \sum_{h=1}^{24} \left(\frac{|p_h^{\text{true}} - p_h^{\text{est}}|}{\bar{p}_{h \text{true}, 24}} \right)^2 - (e_{\text{day}})^2 \quad (31)$$

$$\sigma_{e, \text{week}}^2 = \frac{1}{168} \sum_{h=1}^{168} \left(\frac{|p_h^{\text{true}} - p_h^{\text{est}}|}{\bar{p}_{h \text{true}, 168}} \right)^2 - (e_{\text{week}})^2 \quad (32)$$

3. Case studies

3.1. Introduction

The proposed forecasting models have been applied to predict the electricity prices of the PJM Interconnection, see [PJM \(2003\)](#).

One week per season of year 2002 is studied to validate the performance of the proposed models. The first one corresponds to winter, from February 18 to February 24. For the spring, the selected week is May 20 to May 26, for the summer, August 19 to August 25, and for the fall, November 18 to November 24.

For time series models, the four-step procedure described in Section 2 is carried out. Commercially available software is used to estimate the polynomial parameters of the three models ([SCA, 2003](#)).

To estimate the weighting parameters of the proposed neural network, the solver CONOPT under the GAMS environment is used ([GAMS-CONOPT, 2003](#)). Multiple runs are performed to avoid local minima.

The MATLAB environment (MATLAB (2003)) has been used to implement the wavelet transform models.

The forecasting time framework is the one described in [Fig. 1](#). Therefore, to forecast prices for day d , (i) both price and demand data up to hour 24 of day $d-1$ and (ii) demand predictions for the 24 h of day d are considered known.

3.2. Actual models

The final forecasting models for the PJM Interconnection during year 2002 are described below.

The final ARIMA models for the 4 weeks studied are characterized below. Note that a different model is required for each week. The model for the winter week is:

$$\begin{aligned} & (1 - \phi_1 B^1 - \phi_2 B^2 - \phi_3 B^3)(1 - \phi_{24} B^{24} - \phi_{48} B^{48}) \\ & \times (1 - B^1)(1 - B^{24}) \log(p_h) \\ & = c + (1 - \theta_1 B^1 - \theta_2 B^2 - \theta_3 B^3) \\ & \times (1 - \theta_{24} B^{24} - \theta_{48} B^{48}) \varepsilon_h \end{aligned} \quad (33)$$

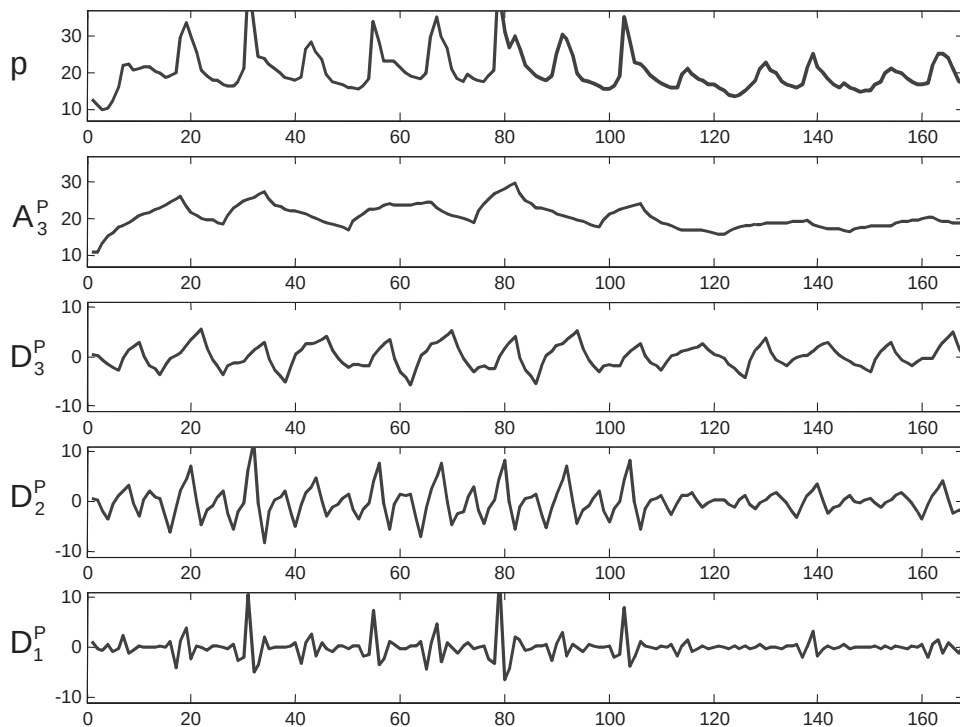


Fig. 7. Wavelet decomposition of the price series for the winter week.

Table 2

PJM Interconnection: price forecasting results for the winter week

Winter week: February 18, 2002 to February 24, 2002

Hours #	True value (\$/MWh)	NAÏVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
M-01	17.41	12.85	26.19	17.14	1.54	17.01	2.32	17.55	0.81	17.12	1.67	18.10	3.95
M-02	16.47	11.03	33.03	16.40	0.44	16.37	0.63	16.15	1.92	15.51	5.80	16.05	2.55
M-03	16.34	10.08	38.31	16.34	0.00	16.19	0.93	15.72	3.80	14.44	11.60	13.89	14.98
M-04	16.47	10.55	35.94	16.73	1.55	16.68	1.27	15.35	6.78	13.91	15.54	14.29	13.25
M-05	17.25	12.53	27.36	18.26	5.84	18.14	5.15	16.08	6.76	14.98	13.16	14.93	13.46
M-06	19.00	15.81	16.79	21.30	12.12	21.00	10.55	17.63	7.19	16.58	12.71	19.43	2.28
M-07	30.35	22.06	27.31	28.06	7.55	27.65	8.91	25.27	16.72	24.07	20.67	21.00	30.81
M-08	21.96	22.30	1.55	29.73	35.39	30.29	37.96	23.20	5.68	23.00	4.77	21.66	1.38
M-09	21.09	20.68	1.94	21.34	1.19	20.72	1.75	21.91	3.91	20.33	3.59	22.34	5.95
M-10	22.33	21.23	4.93	21.77	2.47	21.71	2.77	22.84	2.29	20.86	6.54	20.67	7.44
M-11	20.91	21.50	2.82	21.69	3.71	21.66	3.57	22.27	6.47	20.86	0.23	23.86	14.09
M-12	21.01	21.79	3.71	20.54	2.25	20.46	2.63	21.15	0.66	20.33	3.23	23.26	10.72
M-13	19.26	20.46	6.23	19.31	0.30	19.41	0.77	19.72	2.40	18.72	2.76	20.93	8.69
M-14	18.01	19.99	10.99	18.47	2.56	18.43	2.31	18.85	4.64	18.19	1.00	19.92	10.58
M-15	17.86	18.96	6.16	17.29	3.19	17.22	3.60	17.96	0.56	17.12	4.16	17.44	2.38
M-16	17.04	19.20	12.68	17.17	0.75	17.29	1.48	17.85	4.76	17.12	0.46	17.56	3.05
M-17	18.02	20.08	11.43	17.28	4.12	17.23	4.36	18.70	3.77	17.65	2.01	17.64	2.09
M-18	21.14	29.54	39.74	22.75	7.63	22.80	7.89	25.40	20.14	24.07	13.89	22.95	8.54
M-19	26.16	33.70	28.82	24.22	7.42	24.02	8.19	28.32	8.28	26.75	2.26	29.54	12.91
M-20	24.01	30.02	25.03	24.26	1.03	24.37	1.50	26.71	11.25	25.14	4.73	26.82	11.72
M-21	22.01	25.61	16.36	21.31	3.18	21.36	2.95	24.46	11.11	23.00	4.51	24.67	12.07
M-22	20.06	20.91	4.24	18.95	5.57	18.90	5.82	21.24	5.88	19.79	1.34	19.59	2.36
M-23	17.57	19.29	9.79	16.86	4.02	16.98	3.35	18.29	4.12	17.65	0.51	17.87	1.68
M-24	17.69	18.03	1.92	15.80	10.69	15.97	9.72	17.56	0.77	17.12	3.24	15.76	10.89
W-01	15.60	17.70	13.46	16.75	7.40	16.78	7.58	17.17	10.07	18.19	16.60	17.76	13.84
W-02	15.00	17.00	13.33	15.03	0.18	14.77	1.50	16.13	7.56	17.12	14.13	15.05	0.36
W-03	14.28	16.50	15.55	14.95	4.70	14.78	3.53	15.41	7.89	17.12	19.89	12.49	12.57
W-04	14.31	16.70	16.70	14.51	1.43	14.36	0.38	15.36	7.36	17.12	19.64	12.33	13.84
W-05	15.00	17.10	14.00	15.20	1.36	15.12	0.79	15.98	6.53	18.19	21.27	12.65	15.65
W-06	17.92	20.00	11.61	17.94	0.11	17.66	1.44	18.55	3.51	20.33	13.45	18.10	0.99
W-07	26.12	32.59	24.77	32.11	22.92	31.82	21.80	29.45	12.73	35.31	35.18	22.95	12.12
W-08	27.81	31.23	12.30	24.29	12.68	24.15	13.19	28.00	0.65	29.96	7.72	23.19	16.61
W-09	22.80	26.04	14.21	22.58	0.98	22.99	0.81	24.23	6.27	24.61	7.93	21.65	5.06
W-10	25.00	23.46	6.16	22.89	8.42	22.76	8.96	23.42	6.31	24.61	1.56	19.14	23.44
W-11	24.05	22.79	5.24	24.22	0.72	24.44	1.62	22.59	6.09	23.54	2.12	21.32	11.34
W-12	22.08	21.00	4.89	21.95	0.59	21.97	0.50	20.71	6.22	21.40	3.08	21.87	0.93
W-13	20.00	19.51	2.45	20.87	4.37	20.80	4.00	19.24	3.78	19.79	1.03	20.52	2.58
W-14	20.00	19.00	5.00	19.52	2.40	19.48	2.62	18.60	7.02	19.26	3.70	20.11	0.56
W-15	18.38	18.00	2.07	19.25	4.72	19.30	4.99	17.70	3.68	18.72	1.88	18.25	0.72
W-16	18.00	17.65	1.94	18.10	0.58	18.07	0.36	17.51	2.73	18.19	1.06	18.57	3.19
W-17	18.73	18.35	2.03	19.17	2.35	19.14	2.21	18.22	2.74	19.26	2.83	18.80	0.38
W-18	22.51	21.07	6.40	23.80	5.72	23.87	6.03	23.19	3.01	25.14	11.71	22.29	0.97
W-19	26.51	26.98	1.77	27.35	3.16	27.44	3.53	27.50	3.74	29.96	13.01	27.63	4.23
W-20	24.01	23.80	0.87	23.99	0.07	24.09	0.32	24.42	1.70	26.75	11.41	25.23	5.07
W-21	23.05	21.01	8.85	21.32	7.50	21.26	7.78	21.81	5.38	23.00	0.20	22.79	1.15
W-22	20.00	20.00	0.00	20.17	0.85	20.26	1.29	19.74	1.29	20.33	1.65	18.16	9.19
W-23	18.00	18.00	0.00	17.45	3.04	17.54	2.55	17.89	0.59	18.19	1.06	16.05	10.86
W-24	15.98	17.50	9.51	16.58	3.78	16.79	5.10	17.15	7.31	17.65	10.48	13.67	14.44

(continued on next page)

Table 2 (continued)

Winter week: February 18, 2002 to February 24, 2002

Hours #	True value (\$/MWh)	NAïVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
SA-01	17.01	16.17	4.94	16.67	2.00	16.75	1.51	17.58	3.37	16.58	2.50	17.19	1.04
SA-02	15.78	15.27	3.23	15.79	0.08	15.85	0.43	16.78	6.31	15.51	1.68	14.47	8.32
SA-03	15.20	14.03	7.70	15.29	0.62	15.15	0.30	16.72	9.97	14.44	4.97	13.49	11.23
SA-04	15.20	13.78	9.34	15.11	0.60	15.08	0.79	16.71	9.94	14.44	4.97	13.71	9.80
SA-05	15.46	14.01	9.38	15.78	2.09	15.76	1.93	17.09	10.57	14.44	6.57	14.09	8.87
SA-06	16.15	15.01	7.06	16.47	1.98	16.45	1.88	19.91	23.30	16.05	0.62	16.72	3.51
SA-07	17.51	16.41	6.28	21.67	23.76	21.53	22.97	32.14	83.58	23.54	34.44	16.27	7.08
SA-08	19.03	18.10	4.89	18.51	2.72	18.20	4.40	27.76	45.87	21.93	15.25	16.95	10.95
SA-09	23.44	21.15	9.77	21.19	9.60	20.87	10.95	24.60	4.93	20.33	13.27	19.04	18.76
SA-10	26.02	22.72	12.68	25.70	1.22	25.90	0.47	26.19	0.65	22.47	13.64	18.87	27.48
SA-11	26.00	20.76	20.15	24.45	5.94	24.62	5.29	25.39	2.35	22.47	13.58	21.53	17.19
SA-12	22.02	19.85	9.85	23.55	6.94	23.59	7.15	23.71	7.67	21.40	2.82	20.84	5.37
SA-13	19.00	17.62	7.26	19.57	3.02	19.39	2.08	21.17	11.43	19.26	1.37	18.53	2.46
SA-14	17.54	16.78	4.33	17.96	2.42	17.86	1.83	20.09	14.51	18.19	3.71	16.67	4.99
SA-15	16.40	16.49	0.55	16.40	0.01	16.38	0.09	18.75	14.36	17.12	4.39	15.09	7.96
SA-16	16.02	16.14	0.75	16.16	0.90	16.09	0.42	18.33	14.41	17.12	6.86	15.06	5.99
SA-17	16.40	16.63	1.40	16.59	1.19	16.57	1.04	18.90	15.26	17.12	4.39	15.53	5.31
SA-18	19.24	21.01	9.20	21.10	9.64	20.99	9.09	24.26	26.09	21.93	14.01	19.20	0.22
SA-19	24.04	25.22	4.91	23.03	4.21	22.88	4.82	27.45	14.18	25.14	4.59	24.65	2.53
SA-20	22.98	21.69	5.61	22.51	2.06	22.40	2.55	25.00	8.80	23.00	0.10	22.58	1.72
SA-21	21.7	19.96	8.02	21.49	0.96	21.60	0.44	23.24	7.11	21.93	1.10	20.51	5.49
SA-22	20.01	18.25	8.80	20.14	0.63	20.04	0.14	20.61	3.02	19.79	1.07	17.93	10.41
SA-23	17.8	16.84	5.39	18.48	3.79	18.41	3.41	18.55	4.21	17.65	0.81	17.64	0.90
SA-24	16.8	16.11	4.11	17.01	1.25	16.97	1.00	17.61	4.81	16.58	1.28	15.69	6.63

The ARIMA-generated winter hourly forecast price in (33) depends on previous values of prices as a product of 4 terms: 1 h ago to 3 h ago, 1 day ago to 2 days ago, hourly differentiation, and daily differentiation. It also depends on previous values of errors as a product of 2 terms: 1 h ago to 3 h ago and 1 day ago to 2 days ago.

The model for the spring week is:

$$\begin{aligned}
 & (1 - \phi_1 B^1 - \phi_2 B^2 - \phi_3 B^3)(1 - \phi_{23} B^{23} - \phi_{24} B^{24} \\
 & - \phi_{48} B^{48} - \phi_{72} B^{72} - \phi_{96} B^{96} - \phi_{120} B^{120} \\
 & - \phi_{144} B^{144} - \phi_{168} B^{168})(1 - B^1)(1 - B^{24}) \log(p_h) \\
 & = c + (1 - \theta_1 B^1 - \theta_2 B^2 - \theta_3 B^3)(1 - \theta_{24} B^{24} \\
 & - \theta_{48} B^{48})(1 - \theta_{168} B^{168}) \varepsilon_h
 \end{aligned} \quad (34)$$

The ARIMA-generated spring hourly forecast price in (34) depends on previous values of prices as a product of 4 terms: 1 h ago to 3 h ago, 23 h ago to 1 week ago, hourly differentiation, and daily differentiation.

It also depends on previous values of errors as a product of 3 terms: 1 h ago to 3 h ago, 1 day ago to 2 days ago and 1 week ago.

The model for the summer week is:

$$\begin{aligned}
 & (1 - \phi_{23} B^{23} - \phi_{24} B^{24} - \phi_{48} B^{48} - \phi_{72} B^{72} - \phi_{96} B^{96} \\
 & - \phi_{120} B^{120} - \phi_{144} B^{144} - \phi_{168} B^{168})(1 - B^1) \\
 & \times (1 - B^{24}) \log(p_h) \\
 & = c + (1 - \theta_{23} B^{23} - \theta_{24} B^{24} - \theta_{25} B^{25} - \theta_{48} B^{48}) \\
 & \times (1 - \theta_{168} B^{168}) \varepsilon_h
 \end{aligned} \quad (35)$$

The ARIMA-generated summer hourly forecast price in (35) depends on previous values of prices as a product of 3 terms: 23 h ago to 1 week ago, hourly differentiation, and daily differentiation. It also depends on previous values of errors as a product of 2 terms: 23 h ago to 2 days ago and 1 week ago.

Table 3

PJM Interconnection: price forecasting results for the spring week

Spring week: May 20, 2002 to May 26, 2002

Hours #	True value (\$/MWh)	NAÏVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
M-01	15.47	13.5	12.73	14.45	6.62	14.28	7.69	15.03	2.84	14.93	3.52	15.41	0.40
M-02	14.09	12.85	8.80	13.61	3.43	14.16	0.53	13.63	3.24	12.44	11.73	13.26	5.87
M-03	13.68	11	19.59	13.12	4.07	13.67	0.05	12.81	6.33	12.44	9.08	12.90	5.72
M-04	13.85	10	27.80	13.68	1.21	13.58	1.95	12.31	11.10	12.44	10.20	13.58	1.96
M-05	15.33	12.52	18.33	14.94	2.55	15.03	1.95	12.83	16.30	13.68	10.75	14.03	8.45
M-06	18.21	15.67	13.95	19.14	5.10	18.59	2.10	14.36	21.12	16.17	11.21	17.12	6.01
M-07	24.89	22.45	9.80	25.69	3.20	24.48	1.64	17.74	28.71	21.14	15.05	22.29	10.44
M-08	27.17	29.02	6.81	31.14	14.62	31.66	16.53	21.84	19.63	26.12	3.87	26.35	3.02
M-09	26.7	31.42	17.68	28.91	8.26	29.00	8.61	22.84	14.49	28.61	7.12	29.16	9.23
M-10	31.06	33.64	8.31	29.07	6.41	27.92	10.10	24.01	22.69	29.85	3.90	28.54	8.11
M-11	32.2	36.05	11.96	31.63	1.77	31.43	2.39	25.20	21.75	32.34	0.43	28.84	10.45
M-12	31.24	34.66	10.95	30.76	1.53	31.79	1.77	24.97	20.06	32.34	3.53	28.25	9.56
M-13	28.43	34.09	19.91	30.27	6.45	30.44	7.04	24.74	12.99	32.34	13.73	28.37	0.23
M-14	29.71	34.64	16.59	29.54	0.58	28.94	2.60	25.52	14.11	33.58	13.03	28.10	5.44
M-15	27.8	34.23	23.13	28.67	3.14	28.41	2.20	24.00	13.64	34.83	25.29	27.34	1.66
M-16	27.63	33.92	22.77	26.41	4.44	26.53	3.97	22.98	16.84	33.58	21.54	26.74	3.23
M-17	27.85	33.99	22.05	27.73	0.44	27.44	1.48	22.81	18.08	33.58	20.59	26.62	4.43
M-18	26.16	33.94	29.74	27.30	4.38	27.69	5.88	22.99	12.10	33.58	28.39	26.34	0.68
M-19	23.94	29.06	21.39	24.91	4.04	24.97	4.31	21.57	9.91	31.09	29.87	26.04	8.78
M-20	26.62	29.03	9.05	25.24	5.18	25.01	6.03	23.05	13.39	29.85	12.14	26.25	1.37
M-21	31.61	37.06	17.24	30.37	3.91	30.90	2.23	30.03	4.99	33.58	6.25	28.89	8.60
M-22	26.41	30.8	16.62	26.77	1.37	27.56	4.35	25.65	2.88	29.85	13.03	27.74	5.03
M-23	20.14	20.41	1.34	19.58	2.79	20.61	2.33	19.38	3.78	22.39	11.16	25.34	25.82
M-24	17.46	16.4	6.07	16.74	4.13	16.16	7.45	15.87	9.14	18.66	6.85	20.45	17.14
W-01	16.08	18.09	12.50	17.22	7.10	16.94	5.36	17.19	6.89	17.41	8.29	18.10	12.59
W-02	14.99	17	13.41	14.94	0.33	15.01	0.10	15.70	4.71	16.17	7.87	15.79	5.37
W-03	14	16.36	16.86	15.07	7.64	15.18	8.44	15.23	8.79	14.93	6.61	15.22	8.70
W-04	14	16.46	17.57	14.74	5.27	14.65	4.62	15.04	7.44	14.93	6.61	15.76	12.56
W-05	15.17	17.22	13.51	15.66	3.20	15.31	0.89	15.31	0.92	16.17	6.59	15.98	5.32
W-06	18.31	20.23	10.49	18.93	3.38	18.54	1.28	18.28	0.18	18.66	1.89	19.21	4.92
W-07	26.5	35	32.08	26.61	0.43	26.56	0.23	26.48	0.08	28.61	7.95	24.75	6.59
W-08	28.75	28.11	2.23	27.80	3.31	27.56	4.16	26.19	8.91	28.61	0.50	28.06	2.42
W-09	27.77	27.14	2.27	27.37	1.45	28.13	1.32	25.56	7.96	28.61	3.02	29.27	5.40
W-10	29.64	29.37	0.91	27.75	6.40	28.17	4.96	27.43	7.48	29.85	0.70	28.94	2.38
W-11	29.86	30.37	1.71	30.48	2.07	30.39	1.79	28.70	3.88	32.34	8.30	28.81	3.51
W-12	29.17	29.39	0.75	29.02	0.53	29.43	0.89	27.68	5.13	31.09	6.59	28.35	2.81
W-13	27.69	27.1	2.13	28.26	2.03	28.22	1.90	26.19	5.42	29.85	7.79	27.93	0.85
W-14	28.97	27.12	6.39	28.45	1.80	28.42	1.89	26.89	7.19	29.85	3.05	28.29	2.36
W-15	27.63	26.59	3.76	27.96	1.20	27.97	1.22	25.99	5.95	28.61	3.54	28.12	1.78
W-16	25.02	23.87	4.60	26.70	6.70	26.92	7.57	24.57	1.80	26.12	4.39	28.28	13.01
W-17	27.66	23.91	13.56	26.61	3.80	25.98	6.06	24.30	12.15	24.88	10.06	28.23	2.07
W-18	23.89	22.78	4.65	27.18	13.79	26.90	12.59	23.29	2.51	23.63	1.08	27.52	15.21
W-19	22.98	21.88	4.79	23.25	1.19	23.40	1.82	21.88	4.81	21.14	7.99	25.95	12.93
W-20	23	22.05	4.13	23.87	3.78	23.08	0.35	22.51	2.11	21.14	8.07	25.71	11.79
W-21	31.63	29.38	7.11	29.11	7.98	28.64	9.47	28.81	8.93	26.12	17.43	27.53	12.96
W-22	25.02	24.45	2.28	28.05	12.12	28.10	12.32	23.78	4.96	22.39	10.51	27.58	10.25
W-23	19.45	19.24	1.08	18.99	2.37	20.37	4.72	18.60	4.38	17.41	10.47	25.60	31.63
W-24	17	18.14	6.71	16.79	1.24	16.29	4.21	16.32	4.00	16.17	4.89	19.09	12.31

(continued on next page)

Table 3 (continued)

Spring week: May 20, 2002 to May 26, 2002

Hours #	True value (\$/MWh)	NAïVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
SA-01	14.04	14.92	6.27	15.52	10.51	15.02	6.98	16.08	14.50	16.17	15.15	18.83	34.09
SA-02	12.9	13.97	8.29	13.12	1.74	12.92	0.14	14.30	10.87	14.93	15.70	15.40	19.40
SA-03	11.09	13.33	20.20	12.75	14.97	12.50	12.71	13.46	21.34	13.68	23.37	13.90	25.36
SA-04	10	13.02	30.20	11.54	15.44	11.34	13.44	12.78	27.79	13.68	36.81	12.91	29.08
SA-05	10	13.2	32.00	11.17	11.73	10.78	7.80	13.17	31.68	14.93	49.25	12.44	24.40
SA-06	12.65	13.58	7.35	11.90	5.93	11.41	9.83	14.74	16.48	16.17	27.82	12.42	1.79
SA-07	13.07	14.03	7.35	13.69	4.74	13.46	3.02	17.23	31.83	17.41	33.23	12.25	6.25
SA-08	14.63	18.83	28.71	15.48	5.80	16.00	9.35	20.16	37.76	19.90	36.00	15.68	7.17
SA-09	17	20.43	20.18	17.88	5.17	18.13	6.63	20.94	23.17	19.90	17.06	18.92	11.28
SA-10	18.18	23.29	28.11	20.59	13.24	20.41	12.23	22.35	22.94	21.14	16.29	21.47	18.09
SA-11	19.44	25.8	32.72	20.71	6.53	20.33	4.60	23.96	23.27	23.63	21.57	22.96	18.11
SA-12	19.8	23.62	19.29	20.14	1.74	19.97	0.86	23.67	19.59	23.63	19.37	22.88	15.54
SA-13	19.29	21.78	12.91	19.77	2.50	19.63	1.77	22.81	18.23	23.63	22.49	22.92	18.80
SA-14	19.5	19.96	2.36	20.75	6.42	20.35	4.37	23.30	19.52	23.63	21.21	22.55	15.66
SA-15	19.1	19.88	4.08	19.91	4.29	19.35	1.34	22.11	15.76	23.63	23.76	22.08	15.58
SA-16	18.82	18.83	0.05	19.39	3.00	18.94	0.60	20.79	10.45	22.39	18.94	21.95	16.64
SA-17	18.21	19.02	4.45	19.79	8.67	19.41	6.55	20.34	11.68	22.39	22.93	22.44	23.23
SA-18	18.29	19.4	6.07	19.21	5.02	18.63	1.88	18.95	3.62	22.39	22.42	22.13	20.98
SA-19	18.56	19.71	6.20	18.05	2.72	17.74	4.42	17.36	6.45	21.14	13.94	21.72	17.02
SA-20	18.56	20.05	8.03	18.95	2.10	18.85	1.58	17.48	5.80	21.14	13.94	21.26	14.57
SA-21	20.3	23.15	14.04	22.33	9.97	22.26	9.65	20.53	1.15	23.63	16.41	22.75	12.08
SA-22	19.67	21.79	10.78	20.15	2.43	20.09	2.10	18.62	5.34	22.39	13.80	22.48	14.30
SA-23	17.52	18.74	6.96	15.99	8.75	17.01	2.94	15.30	12.71	17.41	0.63	20.21	15.33
SA-24	14.2	15.75	10.92	14.44	1.67	15.06	6.08	13.59	4.30	16.17	13.87	16.82	18.48

The model for the fall week is:

$$\begin{aligned}
 & (1 - \phi_1 B^1)(1 - \phi_{23} B^{23} - \phi_{24} B^{24} - \phi_{47} B^{47} \\
 & - \phi_{48} B^{48} - \phi_{72} B^{72} - \phi_{96} B^{96} - \phi_{120} B^{120} \\
 & - \phi_{144} B^{144})(1 - \phi_{168} B^{168} - \phi_{169} B^{169})(1 - B^1) \\
 & \times (1 - B^{24}) \log(p_h) = c + (1 - \theta_1 B^1 - \theta_2 B^2) \\
 & \times (1 - \theta_{23} B^{23} - \theta_{24} B^{24} - \theta_{47} B^{47} - \theta_{48} B^{48} \\
 & - \theta_{72} B^{72} - \theta_{96} B^{96} - \theta_{120} B^{120} - \theta_{144} B^{144}) \\
 & \times (1 - \theta_{168} B^{168} - \theta_{169} B^{169} - \theta_{170} B^{170}) \varepsilon_h \quad (36)
 \end{aligned}$$

The ARIMA-generated fall hourly forecast price in (36) depends on previous values of prices as a product of 5 terms: 1 h ago, 23 h ago to 144 h ago, 1 week ago to 169 h ago, hourly differentiation, and daily differentiation. It also depends on previous values of errors as a product of 3 terms: 1 h ago to 2 h ago, 23 h ago to 144 h ago and 1 week ago to 170 h ago.

The final dynamic regression model is:

$$\begin{aligned}
 \log(p_h) = c + & (u_1 B^1 + u_2 B^2 + u_3 B^3 + u_{24} B^{24} \\
 & + u_{25} B^{25} + u_{48} B^{48} + u_{49} B^{49} + u_{72} B^{72} \\
 & + u_{73} B^{73} + u_{96} B^{96} + u_{97} B^{97} + u_{120} B^{120} \\
 & + u_{121} B^{121} + u_{144} B^{144} + u_{145} B^{145} \\
 & + u_{168} B^{168} + u_{169} B^{169} + u_{192} B^{192} \\
 & + u_{193} B^{193}) \log(p_h) + (v_1 B^1 + v_2 B^2 \\
 & + v_3 B^3 + v_{24} B^{24} + v_{25} B^{25} + v_{48} B^{48} \\
 & + v_{49} B^{49} + v_{72} B^{72} + v_{73} B^{73} + v_{96} B^{96} \\
 & + v_{97} B^{97} + v_{120} B^{120} + v_{121} B^{121} + v_{144} B^{144} \\
 & + v_{145} B^{145} + v_{168} B^{168} + v_{169} B^{169} + v_{192} B^{192} \\
 & + v_{193} B^{193}) \log(d_h) + \varepsilon_h \quad (37)
 \end{aligned}$$

The dynamic regression hourly forecast price in (37) depends on previous values of prices from 1 h ago up to 193 h ago. It also depends on previous

Table 4

PJM Interconnection: price forecasting results for the summer week

Summer week: August 19, 2002 to August 25, 2002

Hours #	True value (\$/MWh)	NAïVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
M-01	17.91	18.14	1.28	17.27	3.57	17.35	3.09	16.73	6.59	19.05	6.36	27.28	52.31
M-02	15.87	16.51	4.03	15.12	4.73	14.87	6.25	13.13	17.26	10.58	33.31	23.41	47.49
M-03	15.42	15.51	0.58	14.78	4.15	15.10	2.09	11.96	22.48	10.58	31.40	21.85	41.73
M-04	15.05	15	0.33	15.08	0.20	14.92	0.82	11.10	26.22	10.58	29.68	20.50	36.20
M-05	15.04	15	0.27	15.35	2.05	14.94	0.67	10.88	27.71	10.58	29.67	21.28	41.52
M-06	15.69	16.03	2.17	16.83	7.28	15.77	0.52	11.72	25.30	12.70	19.05	24.25	54.55
M-07	17.2	17.51	1.80	18.92	10.01	18.02	4.76	12.77	25.79	12.70	26.19	28.35	64.80
M-08	22.43	24.59	9.63	23.66	5.46	22.76	1.45	17.20	23.32	19.05	15.09	33.34	48.62
M-09	26.11	28.65	9.73	27.94	7.00	27.88	6.77	19.53	25.21	25.39	2.73	40.07	53.46
M-10	35.02	41.75	19.22	34.00	2.92	34.45	1.65	25.81	26.30	33.86	3.32	45.84	30.89
M-11	48.69	53.55	9.98	48.21	0.98	48.70	0.02	36.04	25.99	44.44	8.73	59.55	22.30
M-12	66.91	56.37	15.75	59.62	10.89	59.00	11.81	44.42	33.60	52.91	20.92	72.88	8.92
M-13	81.74	63.78	21.97	75.09	8.14	75.75	7.32	50.84	37.80	63.49	22.33	82.31	0.70
M-14	72.24	88.34	22.29	92.17	27.58	93.91	30.00	60.64	16.05	76.18	5.46	94.22	30.42
M-15	89.47	97.02	8.44	78.39	12.38	78.08	12.73	68.01	23.99	84.65	5.39	104.44	16.73
M-16	89.31	106.94	19.74	98.12	9.86	100.41	12.43	79.15	11.38	97.35	8.99	115.01	28.77
M-17	94.34	112.04	18.76	91.02	3.51	91.36	3.16	81.82	13.28	103.70	9.92	123.50	30.91
M-18	68.56	106.6	55.48	82.20	19.89	83.12	21.24	72.00	5.02	99.46	45.07	132.49	93.25
M-19	58.42	79.24	35.64	54.96	5.93	55.53	4.95	54.68	6.40	80.42	37.65	99.71	70.68
M-20	49.41	60.93	23.32	50.98	3.17	50.68	2.57	46.35	6.20	67.72	37.05	80.66	63.24
M-21	49.35	58	17.53	48.72	1.28	47.19	4.39	45.04	8.73	65.60	32.93	72.97	47.86
M-22	41.37	52.22	26.23	44.24	6.95	45.20	9.27	40.12	3.01	59.25	43.24	61.14	47.78
M-23	28.69	40.48	41.09	33.20	15.71	33.22	15.77	30.17	5.15	46.56	62.25	53.55	86.64
M-24	17.95	25	39.28	19.69	9.71	18.66	3.96	19.32	7.63	29.63	65.06	46.51	159.11
W-01	23.09	22.51	2.51	17.48	24.30	17.84	22.75	18.69	19.07	16.93	26.68	19.42	15.89
W-02	20.39	18.63	8.63	18.44	9.54	18.66	8.49	15.39	24.54	12.70	37.73	16.62	18.47
W-03	17.25	17.44	1.10	18.26	5.83	18.28	5.95	14.10	18.25	12.70	26.39	13.01	24.61
W-04	15.73	16.7	6.17	15.60	0.77	15.82	0.57	13.11	16.65	10.58	32.72	10.23	34.96
W-05	16.09	16.56	2.92	15.06	6.40	15.13	5.97	13.08	18.68	12.70	21.07	10.82	32.74
W-06	17.14	17.94	4.67	17.05	0.56	16.70	2.60	14.14	17.52	14.81	13.60	13.05	23.86
W-07	16.9	18.99	12.37	18.78	11.12	18.71	10.74	14.80	12.42	14.81	12.32	16.66	1.44
W-08	19.95	20.41	2.31	19.90	0.24	20.17	1.13	18.71	6.19	23.28	16.70	20.20	1.25
W-09	21.61	25.01	15.73	23.69	9.63	23.98	10.97	21.88	1.26	27.51	27.32	25.06	15.97
W-10	25.93	30.54	17.78	26.75	3.14	26.65	2.75	27.12	4.56	35.98	38.72	29.01	11.89
W-11	32.82	37.69	14.84	33.20	1.16	32.83	0.02	34.00	3.58	44.44	35.39	35.71	8.82
W-12	37.64	40.87	8.58	38.22	1.53	38.05	1.08	38.99	3.59	48.67	29.31	42.57	13.11
W-13	43.72	47.46	8.55	43.96	0.56	43.57	0.34	44.92	2.75	55.02	25.86	46.54	6.45
W-14	47.79	50.77	6.24	49.85	4.32	49.39	3.35	50.65	6.00	63.49	32.85	52.69	10.26
W-15	55.43	53.14	4.13	51.90	6.36	52.00	6.19	55.17	0.47	67.72	22.17	58.66	5.83
W-16	59.52	56.85	4.49	60.39	1.46	60.92	2.36	61.93	4.05	74.07	24.44	65.06	9.30
W-17	60.71	64.46	6.18	63.74	4.97	62.18	2.42	65.99	8.69	69.84	15.02	69.98	15.28
W-18	52.51	53	0.93	53.13	1.19	52.71	0.39	57.45	9.41	59.25	12.85	74.98	42.80
W-19	41.85	44.97	7.46	42.18	0.78	42.41	1.33	46.49	11.07	46.56	11.24	55.13	31.74
W-20	35.64	42.25	18.55	36.96	3.71	36.20	1.56	41.21	15.62	38.09	6.88	43.94	23.30
W-21	39.27	42.69	8.71	36.51	7.02	35.11	10.59	41.81	6.49	38.09	2.99	40.33	2.69
W-22	33.15	37.43	12.91	34.22	3.22	34.87	5.17	36.77	10.92	33.86	2.14	34.44	3.90
W-23	21.08	23.06	9.39	23.70	12.44	22.91	8.71	26.70	26.68	21.16	0.40	27.44	30.19
W-24	18.24	18.47	1.26	14.87	18.46	14.69	19.47	19.05	4.46	12.70	30.39	22.35	22.54

(continued on next page)

Table 4 (continued)

Summer week: August 19, 2002 to August 25, 2002

Hours #	True value (\$/MWh)	NAÏVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
SA-01	19.22	28.8	49.84	17.38	9.53	17.57	8.58	17.89	6.91	19.05	0.89	18.68	2.81
SA-02	16.52	18.68	13.08	15.36	7.06	15.07	8.77	14.36	13.11	14.81	10.35	15.25	7.68
SA-03	15.46	18	16.43	15.14	2.11	15.17	1.88	13.33	13.77	14.81	4.18	12.84	16.92
SA-04	14.29	16.13	12.88	14.06	1.63	14.11	1.28	12.43	13.05	12.70	11.17	11.03	22.81
SA-05	14.2	15.86	11.69	13.80	2.84	13.92	1.99	12.34	13.15	14.81	4.29	10.95	22.86
SA-06	14.05	15.83	12.67	14.58	3.75	14.71	4.65	13.18	6.20	14.81	5.40	11.58	17.56
SA-07	14.82	16	7.96	14.12	4.72	14.47	2.35	13.94	5.96	16.93	14.21	12.02	18.92
SA-08	16.95	18.1	6.78	17.99	6.09	17.49	3.16	17.54	3.48	21.16	24.83	13.63	19.61
SA-09	20.48	23.58	15.14	20.72	1.18	20.32	0.76	20.55	0.34	23.28	13.66	17.60	14.08
SA-10	27.93	36.37	30.22	28.28	1.24	28.17	0.84	27.79	0.53	31.74	13.64	21.43	23.26
SA-11	33.7	47.38	40.59	34.34	1.88	33.58	0.35	34.80	3.26	38.09	13.03	23.63	29.89
SA-12	38.85	54.9	41.31	39.32	1.20	38.79	0.16	40.03	3.03	44.44	14.38	25.78	33.64
SA-13	38.22	54.86	43.54	43.00	12.51	41.89	9.60	44.54	16.55	48.67	27.36	25.81	32.47
SA-14	41.96	62.46	48.86	41.84	0.28	40.07	4.49	49.63	18.29	52.91	26.10	27.26	35.04
SA-15	46.93	70.4	50.01	45.13	3.84	44.13	5.97	54.21	15.49	55.02	17.23	29.25	37.67
SA-16	48.59	73.15	50.55	49.82	2.52	48.27	0.65	58.60	20.59	57.14	17.59	31.05	36.11
SA-17	47.38	67.29	42.02	50.91	7.46	50.43	6.45	60.45	27.59	59.25	25.07	32.94	30.47
SA-18	43.53	59.74	37.24	40.71	6.48	40.18	7.68	51.41	18.12	48.67	11.83	34.61	20.50
SA-19	35.61	53.16	49.28	34.75	2.42	35.56	0.14	40.74	14.42	38.09	6.97	30.26	15.03
SA-20	32.75	47.94	46.38	32.32	1.31	32.48	0.81	36.15	10.40	31.74	3.06	28.03	14.41
SA-21	34.22	47.11	37.67	34.23	0.03	33.36	2.51	37.38	9.23	31.74	7.23	26.81	21.66
SA-22	32.18	47.5	47.61	30.11	6.43	30.19	6.18	33.08	2.80	27.51	14.50	23.97	25.52
SA-23	21	35.82	70.57	23.67	12.72	23.48	11.79	23.53	12.03	16.93	19.38	20.11	4.22
SA-24	16.67	23.39	40.31	15.39	7.66	15.10	9.43	16.76	0.54	8.46	49.22	16.12	3.31

values of demands from 1 h ago up to 193 h ago. Note that this model is valid for the whole year.

The final transfer function model is:

$$\begin{aligned}
 \log(p_h) = & c + (w_1 B^1 + w_2 B^2 + w_3 B^3 + w_{24} B^{24} \\
 & + w_{25} B^{25} + w_{48} B^{48} + w_{49} B^{49} + w_{72} B^{72} \\
 & + w_{73} B^{73} + w_{96} B^{96} + w_{97} B^{97} + w_{120} B^{120} \\
 & + w_{121} B^{121} + w_{144} B^{144} + w_{145} B^{145} \\
 & + w_{168} B^{168} + w_{169} B^{169} + w_{192} B^{192} \\
 & + w_{193} B^{193}) \log(d_h) + (1 - \theta_1 B^1 - \theta_2 B^2 \\
 & - \theta_3 B^3 - \theta_{24} B^{24}) (1 - \theta_{168} B^{168}) (1 - \phi_1 B^1 \\
 & - \phi_2 B^2 - \phi_3 B^3 - \phi_{24} B^{24} - \phi_{25} B^{25} \\
 & - \phi_{48} B^{48} - \phi_{49} B^{49} - \phi_{72} B^{72} - \phi_{73} B^{73} \\
 & - \phi_{96} B^{96} - \phi_{97} B^{97} - \phi_{120} B^{120} - \phi_{121} B^{121} \\
 & - \phi_{144} B^{144} - \phi_{145} B^{145} - \phi_{168} B^{168} \\
 & - \phi_{169} B^{169} - \phi_{192} B^{192} - \phi_{193} B^{193})^{-1} \varepsilon_h
 \end{aligned} \quad (38)$$

The transfer function hourly forecast price in (38) depends on previous values of demands from 1 h ago up to 193 h ago. It also depends on a disturbance term that follows an ARMA model, whose numerator is the product of 2 terms: previous values of errors from 1 h ago up to 1 day ago times prices from 1 week ago. The denominator has a single term with previous prices from 1 h up to 193 h ago. Note that this model is valid for the whole year.

The neural network model is characterized by the parameters of the network. These parameters are shown in Table 1 for February 18, 2002. Analogous tables are generated for other days. Nineteen previous prices are required to forecast a given hour h and therefore to evaluate function (9). These prices correspond to $h-1$, $h-2$, $h-3$, $h-24$, $h-25$, $h-48$, $h-49$, $h-72$, $h-73$, $h-96$, $h-97$, $h-120$, $h-121$, $h-144$, $h-145$, $h-168$, $h-169$, $h-192$ and $h-193$.

The wavelet model is characterized by the length of the estimation period, the number of levels considered and the wavelet function selected. In this

Table 5

PJM Interconnection: price forecasting results for the fall week

Fall week: November 18, 2002 to November 24, 2002

Hours #	True value (\$/MWh)	NAÏVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
M-01	16.23	14.21	12.45	14.20	12.48	14.53	10.43	14.89	8.25	14.90	8.20	18.06	11.27
M-02	14.52	12.61	13.15	14.80	1.91	14.95	2.91	14.16	2.49	13.60	6.34	15.38	5.91
M-03	14.31	12.35	13.70	14.13	1.25	14.09	1.52	14.20	0.79	12.95	9.47	13.33	6.88
M-04	14.34	12.5	12.83	14.57	1.63	14.58	1.67	14.59	1.73	12.95	9.68	13.98	2.51
M-05	16.08	13.53	15.86	15.14	5.85	15.11	6.05	14.84	7.72	13.60	15.42	16.50	2.60
M-06	22.03	15.74	28.55	19.55	11.24	18.97	13.89	16.43	25.41	20.73	5.91	25.24	14.58
M-07	31.79	20.85	34.41	29.84	6.12	28.84	9.27	21.52	32.31	36.27	14.10	33.80	6.33
M-08	35.86	25.2	29.73	36.47	1.71	36.44	1.63	24.54	31.57	36.92	2.96	36.63	2.16
M-09	36.14	23.03	36.28	35.94	0.58	35.48	1.83	25.42	29.66	34.97	3.23	35.57	1.57
M-10	36.02	21.88	39.26	37.19	3.27	36.50	1.33	25.61	28.89	34.33	4.69	32.22	10.56
M-11	35.49	23.27	34.43	36.36	2.47	36.24	2.12	25.84	27.17	33.68	5.09	35.89	1.12
M-12	32.79	22.27	32.08	34.10	4.00	33.70	2.79	25.18	23.21	29.79	9.13	36.58	11.55
M-13	30.64	24.05	21.51	31.97	4.32	31.74	3.60	25.12	18.03	27.20	11.22	35.40	15.53
M-14	28.95	23.16	20.00	29.62	2.31	29.40	1.55	24.67	14.79	24.61	15.00	34.70	19.88
M-15	26.7	21.7	18.73	27.56	3.22	27.52	3.05	23.11	13.44	23.32	12.68	31.18	16.77
M-16	26.5	21.82	17.66	26.78	1.08	26.64	0.55	23.25	12.25	23.32	12.01	30.09	13.54
M-17	34.62	27.54	20.45	31.88	7.92	31.91	7.84	30.06	13.18	28.50	17.69	34.72	0.29
M-18	45.59	38.63	15.27	42.58	6.60	43.79	3.94	45.24	0.75	37.57	17.59	45.45	0.31
M-19	45.02	39.57	12.11	40.73	9.54	41.00	8.92	43.50	3.38	31.74	29.50	45.70	1.51
M-20	41.71	30.33	27.28	36.84	11.67	36.25	13.08	34.57	17.13	25.91	37.89	42.75	2.49
M-21	34.59	26.46	23.50	35.92	3.84	35.64	3.03	29.11	15.85	22.02	36.34	38.92	12.52
M-22	29.77	20.2	32.15	28.65	3.78	28.33	4.83	24.15	18.88	17.49	41.26	32.44	8.97
M-23	23.03	18.25	20.76	25.29	9.81	25.84	12.23	21.09	8.41	16.19	29.69	27.75	20.48
M-24	19.34	15.42	20.27	18.63	3.68	18.81	2.75	17.30	10.57	14.25	26.32	22.13	14.41
W-01	15.21	16.63	9.34	17.04	12.06	17.03	11.96	16.56	8.88	16.84	10.74	20.30	33.47
W-02	14.39	15.93	10.70	14.75	2.56	14.82	3.03	16.25	12.94	16.19	12.56	17.29	20.15
W-03	14.12	15.74	11.47	14.81	4.92	14.93	5.76	16.04	13.63	16.19	14.69	15.24	7.95
W-04	14.29	15.65	9.52	14.60	2.18	14.63	2.35	16.14	12.93	16.19	13.28	15.19	6.29
W-05	15.55	17.52	12.67	15.95	2.58	16.08	3.39	18.43	18.51	17.49	12.45	17.86	14.85
W-06	26.56	26.3	0.98	21.99	17.23	21.82	17.87	25.66	3.39	24.61	7.34	26.16	1.49
W-07	40.26	38.31	4.84	38.75	3.76	38.59	4.16	37.11	7.81	36.92	8.30	36.79	8.61
W-08	43.48	39.09	10.10	39.55	9.04	39.73	8.64	38.28	11.95	33.03	24.03	37.99	12.62
W-09	37.7	36.6	2.92	39.37	4.41	40.09	6.34	37.21	1.31	32.38	14.11	35.11	6.87
W-10	36.25	35.47	2.15	35.71	1.49	35.61	1.76	36.85	1.65	31.09	14.23	29.44	18.77
W-11	32.99	35.1	6.40	35.20	6.70	35.86	8.73	37.08	12.43	31.09	5.75	32.34	1.98
W-12	29.57	31.92	7.95	30.54	3.29	30.43	2.92	34.93	18.12	27.20	8.00	32.84	11.04
W-13	25.23	29.07	15.22	27.47	8.88	27.52	9.07	32.19	27.60	23.96	5.02	31.30	24.07
W-14	23.32	27.78	19.13	24.39	4.61	24.38	4.55	30.30	29.95	22.67	2.77	31.28	34.12
W-15	22.55	24.32	7.85	22.09	2.02	22.06	2.18	27.07	20.05	19.43	13.83	27.10	20.16
W-16	22.52	24.58	9.15	23.36	3.74	23.21	3.07	27.48	22.03	20.08	10.83	25.30	12.33
W-17	30.74	32.45	5.56	28.72	6.57	28.35	7.79	36.37	18.33	28.50	7.29	29.03	5.56
W-18	51.62	47.78	7.44	44.30	14.17	45.30	12.25	54.63	5.83	43.39	15.93	39.74	23.01
W-19	44.11	44.14	0.07	46.84	6.19	47.31	7.27	50.89	15.37	39.51	10.43	40.05	9.19
W-20	35.93	36.79	2.39	37.79	5.17	36.81	2.46	42.73	18.94	33.68	6.26	37.65	4.79
W-21	30.23	33.08	9.43	32.96	9.00	32.73	8.25	37.01	22.40	29.15	3.60	35.16	16.32
W-22	22.86	27.16	18.81	25.49	11.49	25.40	11.10	29.02	26.94	22.67	0.83	28.66	25.39
W-23	19.28	20.47	6.17	18.36	4.77	18.32	4.97	23.16	20.12	18.78	2.57	24.60	27.57
W-24	15.67	17.66	12.70	16.34	4.30	16.50	5.31	19.46	24.21	16.84	7.47	19.12	22.05
SA-01	16.7	15.1	9.58	14.37	13.92	14.43	13.61	14.89	10.84	14.90	10.79	19.60	17.38

(continued on next page)

Table 5 (continued)

Fall week: November 18, 2002 to November 24, 2002

Hours #	True value (\$/MWh)	NAïVE		TF		DR		ARIMA		NN		WL	
		Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %	Value	Error %
SA-02	15.69	14.14	9.88	14.96	4.63	15.12	3.60	13.41	14.54	13.60	13.30	15.68	0.06
SA-03	15.33	13.77	10.18	14.79	3.53	14.81	3.38	13.16	14.14	13.60	11.26	14.74	3.87
SA-04	15.28	13.58	11.13	14.72	3.71	14.77	3.38	12.93	15.38	13.60	11.01	15.00	1.82
SA-05	15.51	13.76	11.28	15.40	0.67	15.49	0.10	13.67	11.83	13.60	12.28	16.55	6.70
SA-06	16.51	14.74	10.72	19.23	16.48	19.22	16.42	17.86	8.18	15.54	5.86	20.54	24.41
SA-07	18.39	16.18	12.02	20.81	13.15	20.88	13.50	22.72	23.51	16.84	8.44	20.11	9.34
SA-08	22.39	19.06	14.87	21.63	3.40	21.91	2.14	25.67	14.66	22.02	1.64	22.25	0.62
SA-09	29.96	24.15	19.39	26.69	10.91	26.91	10.20	27.62	7.80	23.96	20.02	26.48	11.63
SA-10	30.93	26.68	13.74	32.12	3.87	32.28	4.38	28.52	7.79	23.32	24.60	26.85	13.20
SA-11	30.49	26.86	11.91	32.10	5.28	31.86	4.50	28.49	6.55	23.32	23.52	30.83	1.12
SA-12	25.87	25.41	1.78	29.53	14.15	29.66	14.67	26.64	2.99	21.37	17.37	30.05	16.18
SA-13	22.19	22.56	1.67	25.16	13.40	25.12	13.19	24.67	11.17	19.43	12.44	27.46	23.76
SA-14	20.9	22.34	6.89	22.31	6.75	22.40	7.18	23.06	10.31	18.14	13.23	25.03	19.75
SA-15	20.46	19.12	6.55	20.38	0.37	20.29	0.81	21.40	4.60	16.84	17.68	21.26	3.90
SA-16	20.67	19.7	4.69	21.60	4.48	21.48	3.92	21.43	3.65	16.84	18.54	20.12	2.66
SA-17	25.24	24.97	1.07	26.48	4.93	26.27	4.08	26.48	4.93	21.37	15.31	25.02	0.88
SA-18	37.9	40.79	7.63	34.49	9.00	34.70	8.45	40.80	7.64	31.74	16.26	35.05	7.52
SA-19	37.6	34.69	7.74	33.30	11.44	34.09	9.35	37.89	0.77	29.79	20.77	35.07	6.73
SA-20	29.43	28.64	2.68	31.03	5.46	30.33	3.05	30.60	3.97	23.32	20.77	32.20	9.40
SA-21	25.83	24.56	4.92	25.37	1.77	24.93	3.50	25.64	0.75	18.78	27.28	29.16	12.88
SA-22	22.6	20.01	11.46	22.44	0.72	22.68	0.35	21.71	3.91	16.19	28.35	24.89	10.15
SA-23	19.58	18.56	5.21	20.86	6.53	20.68	5.61	19.26	1.62	14.25	27.22	22.47	14.73
SA-24	17.01	15.67	7.88	16.75	1.54	16.73	1.61	16.34	3.92	12.31	27.64	17.90	5.26

paper, a 3 level reconstruction technique is applied (see Fig. 7), which decomposes the original series into 3 detail series and 1 approximation series. The trade-off between smoothness and spatial localization has been solved by the use of an order two Daubechies wavelet. An estimation period of two weeks is used because longer estimation intervals originate higher prediction errors due to the inertia of the model.

The naïve forecasting procedure works as follows. The hourly price forecasts for a Monday are the actual hourly prices of the previous Monday. This rule also applies to Saturdays and Sundays. The hourly price forecasts for a Tuesday are the actual hourly prices of the previous day (Monday). This rule also applies to Wednesdays, Thursdays and Fridays.

3.3. Simulation results

Numerical results with the proposed models are presented in Tables 2–11. Tables 2–5 show the

forecast prices for three representative days (Monday, Wednesday and Saturday) of each of the weeks studied. The first column identifies the day and the hour. The second column shows the true market price for each hour in \$/MWh. In the third column, the predicted market price with the naïve procedure for each hour is shown, while in the fourth column, the hourly error is shown. Price and error columns are repeated for each forecasting technique. Instead of pictures that do not provide precise information, the extensive numerical data of Tables 2–5 present actual estimates and actual errors for selected but representative days throughout the year. To perform a detailed analysis this numerical information is more valuable than a pictorial one.

Tables 6–9 summarize some statistical measures that characterize the predictions. These tables contain condensed valuable information for each day of each one of the selected weeks. These numerical data allow an in depth analysis of the

Table 6

PJM Interconnection: statistical measures for the winter week

Winter week: February 18, 2002 to February 24, 2002

Day	NAIVE			TF			DR		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	16.39	0.017	4.203	5.46	0.006	1.900	5.76	0.007	2.011
Tuesday	5.36	0.009	2.284	4.32	0.002	1.316	4.28	0.002	1.263
Wednesday	7.87	0.005	2.115	4.61	0.004	1.617	4.75	0.004	1.602
Thursday	10.75	0.005	2.355	5.17	0.001	1.153	5.14	0.001	1.167
Friday	11.53	0.005	2.837	4.54	0.003	1.525	4.70	0.003	1.543
Saturday	7.34	0.003	1.761	3.81	0.002	1.184	3.71	0.002	1.182
Sunday	5.59	0.002	1.278	3.15	0.001	0.737	3.17	0.001	0.769

Day	ARIMA			NN			WL		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	6.34	0.004	1.770	6.08	0.004	1.808	9.28	0.008	2.609
Tuesday	5.38	0.004	1.766	7.79	0.005	2.200	8.32	0.003	2.100
Wednesday	5.06	0.001	1.210	9.02	0.009	2.635	7.54	0.005	2.121
Thursday	7.83	0.004	1.864	14.07	0.012	3.241	7.07	0.004	1.696
Friday	10.76	0.007	2.817	11.29	0.006	2.833	10.44	0.014	3.237
Saturday	13.80	0.026	4.035	6.82	0.006	1.976	8.15	0.007	2.225
Sunday	4.86	0.002	1.149	6.10	0.002	1.339	5.14	0.002	1.192

Day	NAIVE		TF		DR	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	7.46	8.67	1.90	0.52	1.68	0.78
Tuesday	7.14	1.84	1.23	2.97	0.98	2.67
Wednesday	2.10	5.56	0.54	1.55	0.62	1.29
Thursday	9.61	12.12	1.40	1.84	1.29	2.01
Friday	12.57	10.23	4.78	0.20	4.30	0.09
Saturday	6.84	4.75	0.74	0.97	0.41	0.74
Sunday	7.01	3.80	1.87	0.22	1.96	0.05

Day	ARIMA		NN		WL	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	4.40	0.11	1.22	5.23	2.98	2.94
Tuesday	3.37	4.20	1.53	1.25	1.29	4.51
Wednesday	0.23	1.90	7.63	8.35	5.82	4.16
Thursday	6.13	9.74	13.98	14.17	2.59	4.06
Friday	13.07	7.86	14.32	7.49	13.52	4.34
Saturday	16.57	10.13	1.01	0.32	9.13	5.79
Sunday	5.96	3.32	8.37	3.48	0.83	4.16

behavior of each forecasting method in each of these weeks.

In addition to the 4 weeks under study, Table 10 shows the weekly errors (e_{week}) and their variances ($\sigma_{e,\text{week}}^2$) for the last week (Monday to the last Sunday of the month) of each month of year 2002 in the PJM

Interconnection. Table 11 shows mean square errors for the same weeks. These tables provide a numerical overview of the prediction behavior of each forecasting technique throughout the year. This comprehensive information allows ranking of the considered forecasting techniques.

Table 7

PJM Interconnection: statistical measures for the spring week

Spring week: May 20, 2002 to May 26, 2002

Day	NAIVE			TF			DR		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	15.56	0.006	4.205	4.18	0.001	1.302	4.60	0.002	1.502
Tuesday	10.54	0.006	3.137	6.50	0.006	2.460	7.03	0.009	2.821
Wednesday	6.96	0.005	2.339	4.08	0.001	1.304	4.10	0.001	1.337
Thursday	8.94	0.007	2.668	7.69	0.003	2.015	6.07	0.002	1.696
Friday	9.71	0.008	3.096	5.30	0.002	1.676	4.76	0.002	1.500
Saturday	12.98	0.009	2.643	6.07	0.001	1.153	5.04	0.001	1.004
Sunday	30.29	0.049	5.179	10.24	0.007	1.810	8.07	0.007	1.635

Day	ARIMA			NN			WL		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	14.11	0.007	4.001	12.27	0.008	3.662	6.62	0.003	2.033
Tuesday	12.72	0.006	3.592	8.94	0.008	3.054	10.02	0.009	3.262
Wednesday	5.45	0.001	1.549	6.43	0.002	1.886	7.71	0.003	2.286
Thursday	8.14	0.002	2.051	15.38	0.018	4.419	15.09	0.009	3.886
Friday	13.77	0.015	4.303	15.14	0.012	4.352	10.63	0.011	3.490
Saturday	15.58	0.007	2.915	20.39	0.005	3.539	16.88	0.004	2.949
Sunday	22.42	0.014	3.499	29.14	0.039	4.862	27.09	0.010	4.001

Day	NAIVE		TF		DR	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	13.03	4.60	1.70	1.22	1.70	0.99
Tuesday	2.46	0.39	1.14	0.63	0.90	0.05
Wednesday	0.56	2.05	0.06	2.58	0.13	2.42
Thursday	13.22	3.82	7.37	3.75	5.07	2.63
Friday	4.76	8.69	3.59	2.46	2.88	2.54
Saturday	16.73	9.25	4.91	4.41	4.21	3.27
Sunday	23.09	38.23	1.44	8.38	0.13	6.35

Day	ARIMA		NN		WL	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	16.46	11.37	6.71	7.82	3.45	0.37
Tuesday	6.13	1.54	1.76	4.64	1.74	6.33
Wednesday	5.38	2.32	0.03	0.14	0.64	8.34
Thursday	10.10	0.09	17.01	11.32	15.64	10.67
Friday	10.16	15.64	9.96	13.13	11.19	4.54
Saturday	14.37	10.55	20.52	20.16	14.11	18.58
Sunday	7.74	1.02	26.02	32.57	17.67	36.77

The best results are obtained from the transfer function and dynamic regression models across all seasons. Mean week errors are around 6% and they remain steady for each month, as observed in Table 10. In addition, they use only one model for the whole year 2002, which makes both techniques very robust.

It is specially remarkable that during the summer, the naïve model error is around 25% and those of the dynamic regression and transfer function models are around 6%.

ARIMA and wavelet predictions are less accurate than those of transfer function and dynamic regression.

Table 8

PJM Interconnection: statistical measures for the summer week

Summer week: August 19, 2002 to August 25, 2002

Day	NAIVE			TF			DR		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	20.36	0.042	12.664	9.06	0.012	6.291	9.19	0.165	6.763
Tuesday	32.84	0.120	16.252	6.88	0.004	3.194	6.66	0.004	3.131
Wednesday	7.63	0.003	2.984	4.84	0.002	2.064	4.61	0.002	2.019
Thursday	23.29	0.021	10.933	4.00	0.002	2.425	4.26	0.001	2.323
Friday	17.98	0.018	8.124	7.36	0.002	2.988	6.90	0.002	2.917
Saturday	38.94	0.071	13.470	4.30	0.002	1.704	4.12	0.002	1.643
Sunday	44.11	0.126	15.268	4.92	0.002	1.784	5.31	0.002	1.962

Day	ARIMA			NN			WL		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	17.62	0.029	10.761	21.91	0.032	12.394	41.14	0.100	22.735
Tuesday	21.31	0.037	9.7840	28.38	0.059	12.719	29.66	0.047	12.547
Wednesday	8.82	0.003	3.303	20.38	0.018	7.893	16.04	0.020	6.905
Thursday	11.36	0.011	6.156	19.51	0.046	11.551	11.69	0.011	6.269
Friday	26.86	0.051	12.445	7.99	0.001	3.168	8.73	0.004	3.897
Saturday	11.91	0.014	4.842	15.52	0.014	5.613	24.05	0.036	8.734
Sunday	13.39	0.014	4.854	13.30	0.011	4.585	22.03	0.042	8.130

Day	NAIVE		TF		DR	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	12.81	17.01	0.08	4.34	0.15	14.11
Tuesday	29.70	27.63	3.68	0.30	2.99	1.11
Wednesday	9.80	2.06	1.48	1.24	0.69	1.54
Thursday	24.20	14.14	3.51	0.60	3.81	0.22
Friday	14.97	10.03	4.11	1.52	3.11	0.002
Saturday	36.22	41.57	0.80	0.36	0.23	1.79
Sunday	40.06	47.98	0.03	0.97	0.40	0.98

Day	ARIMA		NN		WL	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	18.93	13.93	7.86	8.15	42.82	39.48
Tuesday	11.29	10.98	20.51	14.00	32.86	26.54
Wednesday	5.31	0.20	17.37	4.35	16.94	2.01
Thursday	15.00	7.14	19.75	18.92	2.72	0.34
Friday	26.63	24.31	2.73	5.86	4.28	5.42
Saturday	8.25	8.98	11.13	5.67	22.98	25.09
Sunday	13.69	11.61	11.73	9.25	19.47	20.25

ssion. ARIMA needs a different model for each of the 12 weeks studied, which makes the model somewhat fragile. Price volatility in spring and specially in the summer makes the ARIMA forecasts much less accurate. However, when the variance of the data is

more stable, as is the case of January and February, then the forecasts are better. Year 2002 shows that ARIMA and wavelet mean week errors are in the 15% range, slightly improving on the naive predictions (17% mean week error).

Table 9

PJM Interconnection: statistical measures for the fall week

Fall week: November 18, 2002 to November 24, 2002

Day	NAIVE			TF			DR		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	24.20	0.017	7.916	5.16	0.002	1.933	5.10	0.002	1.984
Tuesday	6.94	0.002	2.472	5.57	0.002	2.031	5.43	0.001	1.933
Wednesday	7.49	0.002	2.417	6.70	0.003	2.441	6.78	0.003	2.392
Thursday	8.68	0.006	3.027	5.64	0.002	1.906	5.32	0.002	1.780
Friday	6.67	0.007	2.645	4.18	0.002	1.510	4.27	0.002	1.537
Saturday	8.47	0.003	2.339	6.88	0.003	2.004	6.43	0.002	1.884
Sunday	7.11	0.002	1.668	5.59	0.003	1.490	5.78	0.002	1.476

Day	ARIMA			NN			WL		
	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$	$e_{\text{day}} \%$	$\sigma_{e,\text{day}}^2$	$e_{\text{FMSE}}^{\text{day}}$
Monday	16.50	0.016	5.981	16.81	0.022	6.499	7.81	0.004	2.856
Tuesday	13.62	0.005	4.389	8.79	0.002	2.837	12.40	0.007	4.333
Wednesday	14.44	0.006	4.533	10.17	0.007	3.682	14.33	0.009	4.736
Thursday	5.77	0.003	2.047	8.73	0.004	2.754	19.04	0.019	5.997
Friday	7.09	0.002	2.127	6.45	0.003	2.184	14.28	0.009	4.227
Saturday	7.40	0.002	1.982	17.70	0.010	4.662	9.29	0.005	2.654
Sunday	11.92	0.010	3.062	9.06	0.007	2.433	10.46	0.005	2.466

Day	NAIVE		TF		DR	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	27.14	20.04	1.90	1.96	2.46	2.00
Tuesday	0.42	1.50	3.06	3.62	3.16	3.49
Wednesday	0.004	9.78	0.27	0.33	0.64	0.18
Thursday	10.72	4.79	2.63	1.74	2.45	1.81
Friday	5.09	0.03	0.22	1.81	0.91	2.09
Saturday	6.88	6.62	0.41	1.17	0.54	1.20
Sunday	3.32	6.43	1.18	1.98	1.27	2.65

Day	ARIMA		NN		WL	
	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$	$e_{\text{peak}} \%$	$e_{\text{off-peak}} \%$
Monday	19.72	11.76	13.61	17.46	3.27	10.15
Tuesday	10.05	14.31	1.76	8.34	3.61	20.02
Wednesday	7.74	17.46	11.28	0.22	4.65	15.59
Thursday	4.03	2.82	10.20	0.99	6.02	23.00
Friday	2.94	3.69	7.12	4.19	3.37	15.81
Saturday	2.66	2.43	18.36	16.85	1.45	7.18
Sunday	16.90	5.96	9.44	5.50	6.84	2.02

The neural network technique is outperformed by the other methods, including the naïve, and its mean week error is about 19%. This is due to the difficult tune-up of a neural network algorithm, which needs heuristic validation of the model by

means of weight limits adjustments, numbers of hidden layers, numbers of neurons in the input and hidden layers, etc.

With regard to results per season, it is observed that the summer, because of its high volatility and price

spikes, is the most difficult to predict. Only transfer function and dynamic regression models obtain good results in this case, serving as a good benchmark for other models.

4. Conclusions

This paper analyzes different forecasting techniques to predict the 24 market-clearing prices of a day-ahead electric energy market. Price forecasting, both in the short- and the long-term, is required by producers, retailers and consumers to determine their respective bidding strategies in the pool and to engage in beneficial bilateral contracts.

Table 10

Mean week error and variance for the last week of all months of year 2002

Month	NAIVE		TF		DR	
	$e_{\text{week}}^{\text{week}}$ %	$\sigma_{e,\text{week}}^2$	$e_{\text{week}}^{\text{week}}$ %	$\sigma_{e,\text{week}}^2$	$e_{\text{week}}^{\text{week}}$ %	$\sigma_{e,\text{week}}^2$
January	10.60	0.008	5.48	0.004	4.9	0.004
February	9.30	0.008	4.45	0.003	4.52	0.003
March	25.10	0.060	6.59	0.006	6.55	0.005
April	14.22	0.019	9.49	0.010	9.23	0.010
May	12.56	0.011	6.02	0.003	5.53	0.004
June	21.27	0.075	6.89	0.006	6.41	0.005
July	26.11	0.125	5.64	0.003	5.69	0.003
August	25.41	0.060	6.07	0.005	6.03	0.006
September	19.26	0.056	5.05	0.002	4.97	0.002
October	7.45	0.005	5.72	0.003	5.29	0.003
November	10.25	0.011	5.67	0.002	5.58	0.002
December	22.61	0.070	9.72	0.013	7.33	0.008

Month	ARIMA		NN		WL	
	$e_{\text{week}}^{\text{week}}$ %	$\sigma_{e,\text{week}}^2$	$e_{\text{week}}^{\text{week}}$ %	$\sigma_{e,\text{week}}^2$	$e_{\text{week}}^{\text{week}}$ %	$\sigma_{e,\text{week}}^2$
January	6.6	0.006	9.21	0.009	10.81	0.008
February	7.71	0.008	8.73	0.007	8.06	0.007
March	15.34	0.014	17.98	0.027	12.10	0.010
April	15.36	0.023	32.54	0.103	14.89	0.014
May	12.51	0.009	14.21	0.013	12.31	0.008
June	17.53	0.045	27.33	0.057	12.94	0.018
July	24.67	0.084	27.32	0.130	16.64	0.023
August	16.14	0.030	18.45	0.035	22.36	0.061
September	16.82	0.033	15.9	0.030	14.74	0.018
October	10.71	0.010	12.78	0.015	10.64	0.006
November	11.19	0.009	11.15	0.010	12.56	0.010
December	27.77	0.081	32.53	0.116	25.70	0.046

Table 11

Mean square error for the last week of all months of year 2002

$e_{\text{FMSE}}^{\text{week}}$	NAIVE	TF	DR	ARIMA	NN	WL
January	2.814	1.682	1.580	2.065	2.677	2.858
February	2.554	1.392	1.412	2.291	2.369	2.250
March	7.900	2.179	2.230	4.362	5.482	3.548
April	4.948	3.442	3.400	5.407	11.480	4.801
May	3.454	1.729	1.723	3.268	3.797	3.207
June	15.580	4.681	4.418	12.382	16.270	8.402
July	15.640	2.764	2.857	13.553	16.122	7.989
August	12.137	3.272	3.381	8.126	9.052	11.457
September	7.565	1.654	1.637	6.127	5.861	4.972
October	2.702	2.146	2.043	3.913	4.714	3.509
November	3.761	1.926	1.877	3.734	3.851	4.077
December	9.355	4.038	3.156	10.686	12.651	9.016

Extensive testing has been carried out using data from the PJM Interconnection in the year 2002. These data correspond to day-ahead prices, which are demand-weighted averages of locational marginal prices.

Time series techniques reveal themselves as more efficacious than wavelet-transform or neural network techniques. One reason for this is that both wavelet and neural network techniques break the time series of market-clearing prices in some manner.

Among time series techniques, the dynamic regression and transfer function algorithms are more effective than ARIMA models. Wavelet models behave similarly to ARIMA models and neural network procedures do not show good performance. It is remarkable the reasonable results obtained in some cases using the naïve technique.

A promising idea for further research is the combination of wavelet transform and time series algorithms. That is, the historical price series is decomposed using the wavelet transform; the resulting detail and approximation series, which are well behaved, are individually used to forecast their respective future behavior using transfer function algorithms. Then these forecast series (detail and approximation) are used to reconstruct a price forecast series using the inverse wavelet transform algorithm.

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